

MATH 532: ENUMERATIVE COMBINATORICS

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These notes may contain mistakes. Some of the exercises and statements are not provided with a proof. Please use them with caution and verify the information as needed.

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1 Generating Functions

Definition 1.1. Let $\{a_n\}$ be a sequence of complex numbers. The **ordinary generating function** of $\{a_n\}$ is

$$A(x) = \sum a_n x^n$$

Corollary 1.2. Let $\{a_n\}, \{b_n\}, \{c_n\}$ be sequences, $A(x), B(x), C(x)$ be their respective generating functions, then

- (i) $A(x) = B(x) \Leftrightarrow \{a_n\} = \{b_n\}$,
- (ii) if $C(x) = A(x) + B(x)$, then $c_n = a_n + b_n, \forall n$,
- (iii) if $C(x) = A(x)B(x)$, then $c_n = \sum_{k=0}^n a_k b_{n-k}, \forall n$.

Example 1.3. Let $\{F_n\}$ be the Fibonacci Sequence. The ordinary generating function of $\{F_n\}$ is

$$\begin{aligned} F(x) &= 1 + x + (F_1 + F_0)x^2 + \dots \\ &= 1 + x + x(F(x) - 1) + x^2(F(x)). \end{aligned}$$

Solving the equation for $F(x)$ we get

$$\begin{aligned} F(x) &= \frac{1}{1 - x - x^2} \\ &= \frac{1}{(1 - \varphi x)(1 + \varphi^{-1}x)} \\ &= \frac{a}{1 - \varphi x} + \frac{b}{1 + \varphi^{-1}x} \\ &= \sum a\varphi^n x^n + \sum b(-\varphi^{-1})^n x^n \end{aligned}$$

so $F_n = a\varphi^n + b(-\varphi^{-1})^n$, where $\varphi = \frac{1 + \sqrt{5}}{2}$, and we have $a = \frac{1}{\sqrt{5}}, b = \frac{1}{\sqrt{5}}$ by solving the linear fraction of $F(x)$.

Example 1.4. To find $\sum_{k=0}^n \binom{a}{k} \binom{b}{n-k} = \sum_{k=0}^n a_k b_{n-k}$, let

$$\begin{aligned} C(x) &= \sum c_n x^n \\ &= \sum \sum_{k=0}^n a_k b_{n-k} x^n = A(x)B(x) \\ &= (1+x)^a (1+x)^b. \end{aligned}$$

As a result, $c_n = \binom{a+b}{n}$ which implies $\sum_{k=0}^n \binom{a}{k} \binom{b}{n-k} = \binom{a+b}{n}$.

Example 1.5. Let $\{a_n\}$ be a sequence, $a_0 = 1$, and $\sum_{i=0}^n a_i a_{n-i} = 1, \forall n \in \mathbb{N}$.

To find the explicit formula for $\{a_n\}$, we set $\{b_n\} = \{a_n\}$ and $c_n = \sum_{i=0}^n a_i b_{n-i} = 1$, then $C(x) = \sum c_n x^n = \sum x^n = \frac{1}{1-x} = (A(x))^2 \Rightarrow A(x) = \pm(1-x)^{-\frac{1}{2}}$. Based on **Remark 1.6**, we have

$$\begin{aligned} (1-x)^{-\frac{1}{2}} &= \sum \frac{(-\frac{1}{2})(-\frac{1}{2}-1)\dots(-\frac{1}{2}-n+1)}{n!} (-x)^n \\ &= \sum \frac{(\frac{1}{2})(\frac{1}{2}+1)\dots(\frac{1}{2}+n-1)}{n!} x^n \\ &= \sum \frac{1 \cdot 3 \cdot \dots \cdot (2n-1)}{2^n n!} x^n. \end{aligned}$$

$$\text{Thus, } a_n = \frac{1 \cdot 3 \cdot \dots \cdot (2n-1)}{2^n n!} = \frac{1 \cdot 2 \cdot \dots \cdot 2n}{2^n n! \cdot 2 \cdot 4 \cdot \dots \cdot 2n} = \frac{2n!}{2^{2n} (n!)^2} = \frac{1}{2^{2n}} \cdot \binom{2n}{n}$$

Remark 1.6. $\forall \alpha \in \mathbb{R}$, $(1+x)^\alpha$ equals

$$\sum \frac{(\alpha)(\alpha-1)\dots(\alpha-n+1)}{n!} x^n.$$

$$\text{We thus define } \binom{\alpha}{n} = \frac{(\alpha)(\alpha-1)\dots(\alpha-n+1)}{n!}.$$

Definition 1.7. The *exponential generating function* of $\{a_n\}$ is

$$A_e(x) = \sum \frac{a_n}{n!} x^n.$$

Corollary 1.8. Suppose $A_e(x)B_e(x) = \sum \sum_{k=0}^n \frac{a_k}{k!} \frac{b_{n-k}}{(n-k)!} x^n = \sum \frac{c_n}{n!} x^n$, then

$$c_n = \sum_{k=0}^n \binom{n}{k} a_k b_{n-k}.$$

Exercise 1.9. Find the explicit formula for $f(k, n) =$ the number of $\{(X_1, \dots, X_k) \mid X_i \subset \{1, \dots, n\}, X_1 \cap \dots \cap X_k = \emptyset\}$.

From a combinatorial point of view, we know $f(k, n) = (2^k - 1)^n$ since each single number has $2^k - 1$ choices for which subsets X_i they are in.

Proof. Let k be fixed, let $F_k(x) = \sum \frac{f(k, n)}{n!} x^n$.

Consider $X_1, \dots, X_{k-1}$ s.t. $X_1 \cap \dots \cap X_{k-1} = S$, then $X_k \subset \{1, \dots, n\} \setminus S$, so there are $2^{n-|S|}$ ways of choosing X_k . On the other hand, let $Y_i = X_i \setminus S$, then $Y_i \subset \{1, \dots, n\} \setminus S$ and $Y_1 \cap \dots \cap Y_{n-1} = \emptyset$, so we have $f(k-1, n-|S|)$ ways of choosing the Y_i and correspondingly $X_1, \dots, X_{n-1}$.

We thus have

$$\begin{aligned} f(k, n) &= \sum_{S \subset \{1, \dots, n\}} f(k-1, n-|S|) 2^{n-|S|} \\ &= \sum_{s=0}^n \binom{n}{s} f(k-1, n-s) 2^{n-s}. \end{aligned}$$

If we take the whole thing in the summation as b_{n-s} and $a_s = 1$, then we get

$$\begin{aligned}
 F_k(x) &= \sum \frac{x^n}{n!} \sum \frac{f(k-1, m) 2^m x^m}{m!} \\
 &= e^x F_{k-1}(2x) \\
 &= e^x e^{2x} F_{k-2}(4x) \\
 &= \dots = e^{x+2x+\dots+2^{k-1}x} \\
 &= e^{(2^k-1)x} = \sum \frac{(2^k-1)^n x^n}{n!}.
 \end{aligned}$$

We thus have proved that $f(k, n) = (2^k - 1)^n$. □

2 Sets & Multisets

Theorem 2.1. The number of sets $S \subset \{1, 2, \dots, n\}$ with $|S| = k$ is $\binom{n}{k}$.

Corollary 2.2. $\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}$

Example 2.3. Let the number of **strong k -compositions** of n be the number of $\{a_1, \dots, a_k\}$ s.t.

$$a_1 + \dots + a_k = n, a_i > 0.$$

We take the set containing $a_1, a_1 + a_2, \dots, a_1 + \dots + a_k$, clearly it is equivalent to choosing a subset S of $\{1, \dots, n\}$ with $|S| = k$ and $n \in S$. Thus, the number of strong k -compositions of n is $\binom{n-1}{k-1}$.

Similarly, we can prove that the number of **weak k -compositions** of n , which is the number of $\{b_1, \dots, b_k\}$ s.t.

$$a_1 + \dots + a_k = n, a_i \geq 0.$$

is $\binom{n+k-1}{k-1}$ by adding 1 to each b_i and thus form a strong k -composition of $n+k$.

Definition 2.4. A **multiset** is a set allows for multiple instances for each of its elements. The number of multisets S with its elements from $\{1, 2, \dots, n\}$ and $|S| = k$ is denoted as $\left(\binom{n}{k}\right)$, which is equal to $\binom{n+k-1}{n-1} = \binom{n+k-1}{k}$, as such a multiset is equivalent to solving the equation.

Example 2.5. Notice that if we denote the number of appearances of i in S as a_i , then we have

$$a_1 + \dots + a_n = k, a_i \geq 0$$

which is exactly a weak n -composition of k . As a result, the number of weak k -compositions of n is the same as the number of multisets S with its elements from $\{1, 2, \dots, n\}$ and $|S| = k$, which gives

$$\left(\binom{n}{k}\right) = \binom{n+k-1}{n-1} = \binom{n+k-1}{k}.$$

On the other hand, from a generating function point of view, we consider

$$(1 + x_1 + x_1^2 + \dots)(1 + x_2 + x_2^2 + \dots) \dots (1 + x_n + x_n^2 + \dots) = \sum_{b_1, \dots, b_n \geq 0} x^{b_1 + \dots + b_n}.$$

by letting $x_1 = \dots = x_n = x$.

We thus have the above function equal to $\frac{1}{(1-x)^n} = \sum \left(\binom{n}{k}\right) x^k = \sum \binom{n+k-1}{k} x^k$.

Definition 2.6. Let $(y_1 + \dots + y_m)^n = \sum f(a_1, \dots, a_m) y_1^{a_1} y_2^{a_2} \dots y_m^{a_m}$, where $f(a_1, \dots, a_m)$ be the coefficient of $y_1^{a_1} \dots y_m^{a_m}$ for each composition $a_1 + \dots + a_m = n$, then the **multimomial coefficient** is defined as

$$\binom{n}{a_1, \dots, a_m} = f(a_1, \dots, a_m).$$

Theorem 2.7. $\binom{n}{a_1, \dots, a_m} = \frac{n!}{a_1! \dots a_m!}$.

Proof.

$$\begin{aligned}
 (y_1 + \dots + y_m)^n &= [(y_1 + \dots + y_{m-1}) + y_m]^n \\
 &= \sum_{a_m=0}^n \binom{n}{a_m} y_m^{a_m} z^{n-a_m} \\
 &= \sum_{a_m=0}^n \sum_{\sum a_i = n-a_m} \binom{n}{a_m} \binom{n-a_m}{a_1, \dots, a_{m-1}} y_1^{a_1} \dots y_m^{a_m} \\
 &= \sum_{\sum a_i = n} \binom{n}{a_1, \dots, a_m} y_1^{a_1} \dots y_m^{a_m}.
 \end{aligned}$$

So by induction, we have

$$\begin{aligned}
 \binom{n}{a_1, \dots, a_m} &= \binom{n}{a_m} \binom{n-a_m}{a_1, \dots, a_{m-1}} \\
 &= \frac{n!}{a_m!(n-a_m)!} \cdot \frac{(n-a_m)!}{a_{m-1}!(n-a_m-a_{m-1})!} \cdot \dots \cdot \frac{a_1!}{a_1!} \\
 &= \frac{n!}{a_1! \dots a_m!}.
 \end{aligned}$$

□

3 Permutation

Definition 3.1. A *permutation* is a sequence $w = w_1, \dots, w_n$ s.t. $w_i \neq w_j$ if $i \neq j$. It represents a bijective function

$$w : \{1, \dots, n\} \rightarrow \{1, \dots, n\}$$

with $w_i = w(i)$.

Corollary 3.2. $\binom{n}{a_1, \dots, a_m}$ is the number of permutations of a_1 1's, ..., and a_m m 's.

Definition 3.3. Two permutations ω and σ are equivalent if $\exists g \in S_n$ s.t. $\omega = g\sigma g^{-1}$.

$\omega \sim \sigma \Leftrightarrow \omega$ and σ have the same cycle types (number of cycles of different lengths are the same).

Remark 3.4. Notice that from a categorical point of view, if we consider permutations as morphisms from X to X in **FinSet**, then $\omega \sim \sigma$ is basically saying $\exists f : X \rightarrow X$ s.t. $\mathbf{Sym}(f)(\omega) = \sigma$, meaning that the two permutations are structurally the same if we permute the elements of X by f .

Thus, we have the equivalent condition because if we write σ and ω in disjoint cycles then we can achieve σ from ω by permuting their elements iff their cycle types are the same.

Theorem 3.5. The number of $\omega \in S_n$ with cycle type $(c_1, \dots, c_n)$ is

$$\frac{n!}{1^{c_1} c_1! 2^{c_2} c_2! \dots n^{c_n} c_n!}$$

Proof. Giving any permutation of $1, \dots, n$, we have can take the first c_1 of 1 elements as cycles of length 1, c_2 of 2 elements as cycles of length 2, etc. We count the same permutation repetitively, and the number of times we count it as different permutations is

$$1^{c_1} c_1! 2^{c_2} c_2! \dots n^{c_n} c_n!$$

since for each i we have $c_i!$ ways of permuting the cycles of length i and i ways of choosing the first element for each cycle (and the rest will be determined uniquely according to the cycle).

Thus, the same permutation is counted that many times, so the number of different $\omega \in S_n$ with a given cycle type is as claimed. $\square$

Example 3.6. Let $Z_n = \frac{1}{n!} \sum_{w \in S_n} t_1^{c_1(w)} \dots t_n^{c_n(w)}$. Then we have

$$\begin{aligned} Z_n &= \frac{1}{n!} \sum_{(c_1, \dots, c_n), \sum c_i \cdot i = n} \frac{n!}{1^{c_1} c_1! 2^{c_2} c_2! \dots n^{c_n} c_n!} t_1^{c_1} \dots t_n^{c_n} \\ &= \sum_{(c_1, \dots, c_n), \sum c_i \cdot i = n} \frac{1}{1^{c_1} c_1! 2^{c_2} c_2! \dots n^{c_n} c_n!} t_1^{c_1} \dots t_n^{c_n} \end{aligned}$$

We thus have

$$\begin{aligned}
 \sum Z_n x^n &= \sum_{(c_1, \dots, c_n), \sum c_i \cdot i = n} \sum \frac{t_1^{c_1} \dots t_n^{c_n}}{1^{c_1} c_1! 2^{c_2} c_2! \dots n^{c_n} c_n!} x^n \\
 &= \sum \sum \frac{t_1^{c_1} \dots t_n^{c_n}}{1^{c_1} c_1! 2^{c_2} c_2! \dots n^{c_n} c_n!} x^{\sum c_i \cdot i} \\
 &= \sum \frac{(t_1 x)^{c_1} (t_2 x^2)^{c_2} \dots (t_n x^n)^{c_n}}{1^{c_1} c_1! 2^{c_2} c_2! \dots n^{c_n} c_n!} \\
 &= \sum \frac{\left(\frac{t_1 x}{1}\right)^{c_1}}{c_1!} \dots \frac{\left(\frac{t_n x^n}{n}\right)^{c_n}}{c_n!} \dots \\
 &= \sum \frac{\left(\frac{t_1 x}{1}\right)^{c_1}}{c_1!} \sum \frac{\left(\frac{t_2 x^2}{2}\right)^{c_2}}{c_2!} \dots \sum \frac{\left(\frac{t_n x^n}{n}\right)^{c_n}}{c_n!} \dots \\
 &= \exp \frac{t_1 x}{1} \dots \exp \frac{t_n x^n}{n} \dots \\
 &= \exp \frac{t_1 x}{1} + \dots + \frac{t_n x^n}{n}
 \end{aligned}$$

Remark 3.7. We used a very common trick in the above reasoning:

$$\sum_{n=0}^{\infty} \sum_{a+b=n} f(a, b) = \sum_{(a, b)} f(a, b).$$

Example 3.8. We want to compute the number a_n of $w \in S_n$ s.t. $w^2 = id$. Apparently, in such w every cycle has length 1 or 2. In other words, $c_i = 0$ if $i \geq 3$.

$$\text{Thus, } Z_n = \frac{1}{n!} \sum_{w \in S_n, w^2 = id} t_1^{c_1(w)} t_2^{c_2(w)} = \frac{1}{n!} \sum 1 = \frac{1}{n!} a_n \text{ by letting } t_1 = t_2 = 1.$$

$$\text{Thus, } \sum a_n \frac{x^n}{n!} = \sum Z_n(1, 1, 0, \dots) x^n = \exp\left(x + \frac{x^2}{2}\right).$$

By letting $t_i = 1$, we are going to use the following lemma frequently:

Lemma 3.9. $\sum \frac{x^n}{n} = \ln \frac{1}{1-x}.$

Proof. Let $\sum \frac{x^n}{n} = f(x)$, then we have

$$\begin{aligned}
 \frac{d}{dx} f(x) &= 1 + \frac{2x}{2} + \dots \\
 &= \frac{1}{1-x},
 \end{aligned}$$

which tells us $f(x) = \int_0^x \frac{1}{1-y} dy = \ln \frac{1}{1-x}.$ □

Example 3.10. We want to compute $E_k(n) = \mathbb{E}[c_k(w)]$ where $w \in S_n$, which refers to the average number of k -cycle in a permutation of n elements.

We have $\mathbb{E}[c_k(w)] = \frac{1}{n!} \sum c_k(w)$. Taking the derivative of Z_n in regard to t_k , we have

$$\begin{aligned}
 \frac{\partial Z_n}{\partial t_k} &= \frac{1}{n!} \sum c_k(w) t_1^{c_1(w)} \dots t_k^{c_k(w)-1} \dots \\
 &= \frac{1}{n!} \sum c_k(w)
 \end{aligned}$$

by letting $t_i = 1$.

We also have

$$\begin{aligned}
 (1) \quad \frac{\partial}{\partial t_k} (\sum Z_n x^n) \Big|_{t_i=1} &= \frac{\partial}{\partial t_k} (\exp(t_1 x + \dots)) \Big|_{t_i=1} \\
 (2) \quad &= \frac{x^k}{k} \exp(x + \frac{x^2}{2} + \dots) \\
 (3) \quad &= \frac{x^k}{k} \frac{1}{1-x} \\
 (4) \quad &= \frac{1}{k} \sum_{n=k} x^n.
 \end{aligned}$$

Thus, $E_k(n) = \frac{1}{k}$.

Remark 3.11.

$$\begin{aligned}
 E_k(n) = \frac{1}{k} &\Leftrightarrow k \sum_{w \in S_n} c_k(w) = n! \\
 &\Leftrightarrow \sum_{w \in S_n} k \cdot c_k(w) = n! \\
 &\Leftrightarrow \sum_{w \in S_n} \sum_{i=1}^n 1(w_i) = n!, \text{ where } 1(w_i) = 1 \text{ iff } w_i \text{ is in a } k\text{-cycle.} \\
 &\Leftrightarrow \sum_{i=1}^n \sum_{w \in S_n} 1(w_i) = n! \\
 &\Leftrightarrow \sum_{i=1}^n \sum_{w \in S_n} 1 = n!, \text{ where } w_i \text{ is in a } k\text{-cycle of } w.
 \end{aligned}$$

We then know such w has a structure of $(w_i, a_1, \dots, a_{k-1})(\dots)$, so there are in total

$$\binom{n-1}{k-1} \cdot (k-1)! \cdot (n-k)! = (n-1)!$$

since we have $\binom{n-1}{k-1}$ ways to choose the $k-1$ elements in the k -cycle containing w_i , $(k-1)!$ ways of permuting them, and $(n-k)!$ ways of permuting the rest elements which uniquely determine the cycle structure of the rest of w .

Therefore, $\sum_{i=0}^n \sum_{w \in S_n} 1 = \sum_{i=1}^n (n-1)! = n!$, so $E_k(n) = \frac{1}{k}$.

Definition 3.12. $C(n, k)$, defined as the number of $w \in S_n$ with a total of k cycles, is called the **Stirling's number of the first kind**.

Theorem 3.13. $C(n, k) = C(n-1, k-1) + (n-1)C(n-1, k)$.

Proof. Suppose (n) is a cycle itself, then there are $C(n-1, k-1)$ ways of arranging the rest $n-1$ elements. On the other hand, if n is in one of the k cycles then there are $n-1$ choices for where to put it and $C(n-1, k)$ ways of arranging the rest $n-1$ elements. $\square$

Theorem 3.14. $\sum_{k=1}^n C(n, k) t^k = t(t+1)\dots(t+n-1)$.

Proof. **1st Proof:** We induct on n .

$$\begin{aligned}
 \sum_{k=1}^{n+1} C(n+1, k)t^k &= \sum_{k=1}^{n+1} C(n, k-1)t^k + \sum_{k=1}^{n+1} nC(n, k)t^k \\
 &= tt(t+1)\dots(t+n-1) + nt(t+1)\dots(t+n-1) \\
 &= (n+t)t(t+1)\dots(t+n-1) = t(t+1)\dots(t+n).
 \end{aligned}$$

2nd Proof: We have $Z_n = \frac{1}{n!} \sum t^{c_1(w)+\dots+c_n(w)} = \frac{1}{n!} \sum_{k=1}^n C(n, k)t^k$ by letting $t_i = t$.

Thus,

$$\begin{aligned}
 \sum \frac{x^n}{n!} \sum C(n, k)t^k &= \sum Z_n x^n \\
 &= \exp\left(t\left(x + \frac{x^2}{2} + \dots\right)\right) \\
 &= \exp\left(t \cdot \log \frac{1}{1-x}\right) \\
 &= \left(\frac{1}{1-x}\right)^t = \frac{1}{(1-x)^t} \\
 &= \sum \binom{t+n-1}{n} x^n.
 \end{aligned}$$

Thus, $\sum C(n, k)t^k = \binom{t+n-1}{n} \cdot n! = t(t+1)\dots(t+n-1)$. □

4 Inversion

Definition 4.1. Let $w = w_1 \dots w_n$. (i, j) is an **inversion** of w if $i < j$ and $w_i > w_j$.

$$\text{inv}(w) = |\{(i, j) \mid i < j, w_i > w_j\}|, \text{ so we have } \text{inv}(w) \leq \binom{n}{2}.$$

Corollary 4.2. Let $s_i = (i, i+1)$, then $\text{inv}(w \circ s_i) = \text{inv}(w) \pm 1$.

Theorem 4.3.

$$\begin{aligned} \sum_{w \in S_n} q^{\text{inv}(w)} &= 1 + (n-1)q + \dots + q^{\binom{n}{2}} \\ &= 1(1+q)(1+q+q^2)\dots(1+q+\dots+q^{n-1}) \\ &= [1]_q \cdot \dots \cdot [n]_q = [n]_q! \end{aligned}$$

where $[a]_q = 1 + q + \dots + q^{a-1} = \frac{q^a - 1}{q - 1}$.

Proof. 1st Proof: We induct on n .

$$\begin{aligned} \sum_{w \in S_{n+1}} q^{\text{inv}(w)} &= \sum_{j=1}^{n+1} \sum_{w \in S_n} q^{\text{inv}(w) + n+1-j} \\ &= \sum_{j=1}^{n+1} q^{n+1-j} \sum_{w \in S_n} q^{\text{inv}(w)} \\ &= \sum_{j=1}^{n+1} q^{n+1-j} \cdot [n]_q! \\ &= [n+1]_q!. \end{aligned}$$

The first step is due to the following remark. □

Remark 4.4. Assume $w = \dots(n+1)\dots$, where $n+1$ is at the j th position, then $n+1$ creates $n+1-j$ inversions. The remaining number of inversions is exactly $\text{inv}(w_1 \dots w_{j-1} w_{j+1} \dots w_n)$. Therefore, $\text{inv}(w) = n+1-j + \text{inv}(w_1 \dots w_{j-1} w_{j+1} \dots w_n)$.

Definition 4.5. The **code** of w is defined as $c_i = |\{w^{-1}(i) < w^{-1}(j), j < i\}|$.

Remark 4.6. The sequence of codes $c_1, \dots, c_n$ uniquely determines a permutation if we determine the initial positions of each elements from back.

Corollary 4.7. $\sum_{i=1}^n c_i = \text{inv}(w)$.

Proof. 2nd Proof: We have

$$\begin{aligned} \sum_{w \in S_n} q^{\text{inv}(w)} &= \sum_{c_i \in [0, i-1]} q^{c_1 + \dots + c_n} \\ &= \prod_{i=1}^n \left(\sum_{c_i=0}^{i-1} q^{c_i} \right) \\ &= \prod_{i=1}^n (1 + \dots + q^{i-1}) \\ &= \prod_{i=1}^n [i]_q = [n]_q!. \end{aligned}$$

□

5 Descent

Definition 5.1. The **descent** of w is defined as $D(w) = \{i : w_i > w_{i+1}\}$.

We also define $\text{maj}(w) = \sum_{i \in D(w)} i$.

Theorem 5.2. $\sum_{w \in S_n} q^{\text{maj}(w)} = [n]_q! = \sum_{w \in S_n} q^{\text{inv}(w)}$.

Remark 5.3. To prove this theorem, we have a bijection named Foata's Bijection, which is a function $\varphi : w \rightarrow \varphi(w)$ s.t. $\text{maj}(w) = \text{inv}(\varphi(w))$.

For k from 1 to n :

$\gamma_k \rightarrow$ a permutation of $w_1 \dots w_k$.

γ_k is mapped to a permutation of $w_1 \dots w_k$ and γ_{k+1} is determined by γ_k by the following algorithm.

If $w_k > w_{k+1}$, then γ_{k+1} is generated by separating $\gamma_k * w_{k+1}$ by straight lines after elements that are greater than w_{k+1} . Then in each compartment, we move the last element (the one before the straight line) to the first place and everything else moved right for one position. If $w_k < w_{k+1}$, then [greater than] is replaced with [less than] in the algorithm.

Let $\varphi_w = \gamma_n$, then $\text{inv}(\gamma_n) = \text{maj}(w)$.

Definition 5.4. $\text{des}(w)$ is defined as the number of descents in w . Its value is $|D(w)|$.

Definition 5.5. The **Eulerian polynomial** $A_d(x) = \sum_{w \in S_d} x^{1+\text{des}(w)} = \sum_{k=1}^d A(d, k) x^k$. where $A(d, k)$ is the number of permutations $w \in S_d$ with exactly $k-1$ descents, named the **Eulerian number**.

Theorem 5.6. $A(d+1, k) = kA(d, k) + (d+2-k)A(d, k-1)$

Proof. We prove the statement by induction. Suppose the statement holds for d , then we first let $\bar{w} = w_1 \dots w_i w_{i+2} \dots w_{d+1}$ be a permutation in S_d , with i arbitrary and $d+1$ inserted at w_{i+1} .

If $w_i > w_{i+2}$, then $d+1$ does not affect the number of descents in w and there are k places to choose $i+1$: after each descent or in the end. Therefore, the number for this case equals $kA(d, k)$, since $\bar{w}$ needs to have $k-1$ descents.

On the other hand, if $w_i < w_{i+2}$, then $d+1$ increased the number of descents by 1 so there are $k-2$ descents in $\bar{w}$. There are $d-k+2$ places to choose $i+1$: after $d-k+1$ that is not a descent (or the last) or the beginning. Therefore, the number for this case equals $(d+2-k)A(d, k-1)$, since $\bar{w}$ needs to have $k-2$ descents.. The recursive formula is therefore attained. $\square$

Theorem 5.7. For every $d \geq 0$, we have

$$\sum_{m \geq 0} m^d x^m = \frac{A_d(x)}{(1-x)^{d+1}}.$$

Proof. We prove by induction on d . When $d=0$, $\frac{1}{1-x} = \sum x^m$ so the base case holds. Suppose the claim holds for d , then consider $d+1$.

WTS $\sum_{m \geq 0} m^{d+1} x^m = \frac{A_{d+1}(x)}{(1-x)^{d+2}}$. Differentiate $\sum m^d x^m$ and multiply both sides by x we get $\sum m^{d+1} x^m = \frac{x(1-x)A'_d(x) + (d+1)xA_d(x)}{(1-x)^{d+2}}$. Thus it suffices to show that $A_{d+1}(x) = x(1-x)A'_d(x) + (d+1)xA_d(x)$.

We have RHS equals to

$$\begin{aligned}
 x(1-x)A'_d(x) + (d+1)xA_d(x) &= x(1-x) \sum kA(d,k)x^{k-1} + (d+1) \sum A(d,k)x^{k+1} \\
 &= \sum k(1-x)A(d,k)x^k + \sum (d+1)A(d,k-1)x^k \\
 &= \sum kA(d,k)x^k - \sum (k-1)A(d,k-1)x^k + \sum (d+1)A(d,k-1)x^k \\
 &= \sum kA(d,k)x^k + \sum (d+2-k)A(d,k-1)x^k \\
 &= \sum A(d+1,k)x^k,
 \end{aligned}$$

which is exactly the expression of LHS. $\square$

Corollary 5.8. $\sum A_d(x) \frac{t^d}{d!} = \frac{1-x}{1-xe^{(1-x)t}}$

Proof. We have

$$\begin{aligned}
 \sum_{d \geq 0} \frac{A_d(x)}{(1-x)^{d+1}} \frac{t^d}{d!} &= \sum_{d \geq 0} \sum_{m \geq 0} m^d x^m \frac{t^d}{d!} \\
 &= \sum_{m \geq 0} \sum_{d \geq 0} m^d x^m \frac{t^d}{d!} \\
 &= \sum_{m \geq 0} x^m e^{mt} \\
 &= \frac{1}{1-xe^t}.
 \end{aligned}$$

Multiplying both sides by $\frac{1}{(1-x)}$ and substitute $(1-x)t$ for t we get the wanted equality. $\square$

Definition 5.9. A map f is **w-compatible** if

- (i) $f(w_1) \geq \dots \geq f(w_d)$
- (ii) $f(w_i) > f(w_{i+1})$ if $w_i > w_{i+1}$.

Corollary 5.10. For any $f : [d] \rightarrow \mathbb{N}$, $\exists! w$ s.t. f is w-compatible.

Example 5.11. Let $f : [3] \rightarrow \mathbb{N}$ with $f(1) = 3$, $f(2) = 4$, $f(3) = 2$. Then $w = 213$ makes f w-compatible.

Basically, if $f([d]) = \{a_1, \dots, a_m\}$ after being ordered, then $w = f^{-1}(a_m) \dots f^{-1}(a_1)$ with each pre-image ordered from least to the greatest.

Corollary 5.12. Let $\mathcal{A}_m(w) = \{f : [d] \rightarrow [m] \mid f \text{ w-compatible}\}$, then we have

$$m^d = \sum_{w \in S_d} |\mathcal{A}_m(w)|$$

Lemma 5.13. $|\mathcal{A}_m(w)| = \binom{m+d-1-\text{des}(w)}{d} = \left(\binom{m-\text{des}(w)}{d} \right)$

Proof. Let $w = w_1 < w_2 < \dots < w_{i_1} > w_{i_1+1} < \dots$. In other words, w has a descent at w_{i_k} . Then we have

$$m \geq f(w_1) \geq \dots \geq f(w_{i_1}) > f(w_{i_1+1}) \geq \dots$$

Taking $y_i = f(w_i)$. we have

$$m \geq y_1 \geq \dots \geq y_{i_1} > y_{i_1+1} \geq \dots$$

The number of $>$ is $\text{des}(w)$ since f is w -compatible.

Therefore, the number of such $\{y_1, \dots, y_d\}$ for a given w is $\binom{m+d-1-\text{des}(w)}{d}$. Thus, we have $m^d = \sum_w \binom{m+d-1-\text{des}(w)}{d} = \sum_{k=0}^{d-1} A(d, k+1) \binom{m+d-1-k}{d}$.

Thus, we have $\sum m^d x^m = \sum_{k=0}^{d-1} A(d, k+1) \binom{m+d-1-k}{d} x^m$. □

Corollary 5.14. $\sum m^d x^m = \sum_{k=0}^{d-1} A(d, k+1) \binom{m+d-1-k}{d} x^m = \frac{A_d(x)}{(1-x)^{d+1}}$

Theorem 5.15. $\sum_{f:[n] \rightarrow \mathbb{N}, f w\text{-compatible}} q^{f(1)+\dots+f(n)} = \frac{q^{\text{maj}(w)}}{(1-q)(1-q^2)\dots(1-q^n)}$

6 Partition

Definition 6.1. $\lambda = (\lambda_1, \dots, \lambda_k)$ is a **partition** of n if $\lambda_1 \geq \dots \geq \lambda_k > 0$, $\lambda_1 + \dots + \lambda_k = n$, denoted as $\lambda \vdash n$.

$p(n)$ is the number of integer partitions of n .

$p_k(n)$ is the number of partitions of n with exactly k parts.

$p_{\leq k}(n)$ is the number of partitions of n with $\leq k$ parts.

$l(\lambda)$ is the number of parts in λ .

The **Young diagram** of a partition is formed by putting λ_i squares on the i th row. By rotating λ 's young diagram, we get a conjugate diagram, denoted as λ' (i.e. λ_i squares in the i th column).

Corollary 6.2. $l(\lambda') = \lambda_1$

Theorem 6.3. $\#\{\lambda \vdash n \mid \lambda = \lambda'\} = \#\{\lambda \vdash n \mid \lambda_1 > \dots > \lambda_k, \lambda_i \text{ odd}\}$

Proof. We can form a bijection between the two set by letting $\bar{\lambda} = (2\lambda_1 - 1, \dots, 2\lambda_i - 2i + 1)$ if $\lambda = (\lambda_1, \dots, \lambda_k)$. $\square$

Theorem 6.4. $\sum_n p_{\leq k}(n)q^n = \frac{1}{(1-q)\dots(1-q^k)}$

Proof. There is clearly a combinatorial proof by mapping each partition of n with less than k parts with a partition of n with every part less than or equal to k (mapping λ to λ'). Since the RHS is $\sum q^{n_1} \sum (q^2)^{n_2} \dots \sum (q^k)^{n_k}$, we have the RHS equal to partition of n with every part less than or equal to k .

To prove the statement with generating function, we have:

$$\begin{aligned} \sum_n p_{\leq k}(n)q^n &= \sum_{\lambda_1 \geq \dots \geq \lambda_k \geq 0} q^{\lambda_1 + \dots + \lambda_k} \\ &= \sum_{m_1, \dots, m_k} q^{km_k + \dots + 1m_1} \\ &= \sum q^{m_1} \dots \sum (q^k)^{m_k} \\ &= \frac{1}{(1-q)\dots(1-q^k)} \end{aligned}$$

where m_i denotes the number of columns in the Young diagram of λ with exactly i squares.

Notice that we have $k \rightarrow \infty \Rightarrow \sum p(n)q^n = \prod_{k=1}^{\infty} \frac{1}{(1-q^k)}$. $\square$

Theorem 6.5. Let $q(n) = \#\{\lambda \vdash n \mid \lambda_i > \lambda_{i+1}\}$ and $p_0(n) = \#\{\lambda \vdash n \mid \lambda_i \text{ odd}\}$, then $q(n) = p_0(n)$.

First Proof. WTS $\sum q(n)t^n = \sum p_0(n)t^n$. We have

$$\sum_{\lambda_1 > \dots} t^{\lambda_1 + \dots} = (1+t)(1+t^2)\dots$$

and

$$\begin{aligned}
 \sum p_0(n)t^n &= \sum_{\substack{\lambda_1 \geq \dots \\ \lambda_i \text{ odd}}} t^{\lambda_1 + \dots} \\
 &= \sum_{m_i \in \mathbb{Z}} t^{1m_1 + 3m_3 + \dots} \\
 &= \sum t^{m_1} \sum (t^3)^{m_3} \dots \\
 &= \prod_{k=1}^{\infty} \frac{1}{1 - t^{2k-1}}
 \end{aligned}$$

where m_i is the number of λ_j with $\lambda_j = i$.

Therefore, the statement is equivalent to:

$$\begin{aligned}
 (1+t)(1+t^2)\dots &= \prod_{k=1}^{\infty} \frac{1}{1 - t^{2k-1}} \\
 \Leftrightarrow (1+t)(1+t^2)\dots &= \frac{(1-t^2)(1-t^4)\dots}{(1-t)(1-t^2)(1-t^3)\dots} \\
 &= \frac{(1+t)(1-t)(1+t^2)(1-t^2)}{(1-t)(1-t^2)(1-t^3)\dots} = (1+t)(1+t^2)\dots
 \end{aligned}$$

□

Second Proof. Let $\lambda = (7, 7, 7, 7, 5, 5, 3, 3, 3, 1, 1)$. Based on the number of appearance of each number, we denote λ as $(7^4, 5^2, 3^3, 1^2)$.

Then, we denote the exponent in binary form. In this example, $4 = 2^2$, $2 = 2^1$, $3 = 2^1 + 2^0$, and $2 = 2^1$. We then multiply each part by the base and we get $\{2^2 \cdot 7, 2^1 \cdot 5, 2^1 \cdot 3, 2^0 \cdot 3, 2^1 \cdot 1\}$.

Sorting this set we get $\bar{\lambda} = \{28, 10, 6, 3, 2\}$, which is a partition of n with all parts distinct.

To go back, we express $\bar{\lambda}_i = 2^{n_i} k$ where k is odd. Then based on the sum of coefficient in front of k ($\sum n_i$ if $\bar{\lambda}_i = 2^{n_i} k$), we get the number of appearances of k in λ . Here, $\bar{\lambda} = \{2^2 \cdot 7, 2^1 \cdot 5, 2^1 \cdot 3, 2^0 \cdot 3, 2^1 \cdot 1\}$ so we get 4 of 7, 2 of 5, 3 of 3, and 2 of 1 in λ . □

Theorem 6.6. $\prod_{i=1}^{\infty} \frac{1}{1 - xq^i} = \sum_k \frac{x^k q^k}{(1-q)\dots(1-q^k)}$

Proof.

$$\begin{aligned}
 \prod_{i=1}^{\infty} \frac{1}{1-xq^i} &= \sum_{m_1} (xq)^{m_1} \sum_{m_2} (xq^2)^{m_2} \dots \\
 &= \sum_{m_1, m_2, \dots} x^{m_1+\dots} q^{1m_1+2m_2+\dots} \\
 &= \sum_{\lambda} x^{l(\lambda)} q^{|\lambda|} \\
 &= \sum_k \sum_{\lambda, l(\lambda)=k} x^k q^{|\lambda|} \\
 &= \sum_k \sum_{\lambda, l(\lambda)=k} x^k q^{k+|\mu|} \\
 &= \sum_k (xq)^k \sum_{l(\mu) \leq k} q^{|\mu|} \\
 &= \sum_k (xq)^k \frac{1}{(1-q)\dots(1-q^k)}
 \end{aligned}$$

where μ is λ without the first column. The last step is a direct result of **Theorem 6.4**. $\square$

Theorem 6.7. $\prod_{k=1}^{\infty} (1-x^k) = 1 + \sum_{m \geq 1} (-1)^m (x^{\frac{m(3m-1)}{2}} + x^{\frac{m(3m+1)}{2}})$

Proof.

$$\begin{aligned}
 \prod_{k=1}^{\infty} (1-x^k) &= \sum_{s=\{s_1 < \dots < s_n\}} (-1)^n x^{s_1+\dots+s_n} \\
 &= \sum_{\lambda, \lambda_i > \lambda_{i+1}} (-1)^{l(\lambda)} x^{|\lambda|} \\
 &= \sum_n x^n \left(\sum_{\substack{\lambda \vdash n, l(\lambda) \text{ even}, \\ \lambda_i > \lambda_{i+1}}} 1 - \sum_{\substack{\lambda \vdash n, l(\lambda) \text{ odd}, \\ \lambda_i > \lambda_{i+1}}} 1 \right)
 \end{aligned}$$

Now suppose $\lambda = (6542)$, we call the last row of λ as λ_{last} and the rightmost diagonal $diag(\lambda)$.

Suppose λ_{last} and $diag(\lambda)$ do not overlap or $\lambda_{last} - diag(\lambda) \neq 0$ or 1.

If $\lambda_{last} < diag(\lambda)$ then we can get a new partition μ with $|\mu| = |\lambda|$ and $l(\mu) = l(\lambda) - 1$ by moving the λ_{last} to the right of $diag(\lambda)$.

If $\lambda_{last} > diag(\lambda)$ then we can get a new partition μ with $|\mu| = |\lambda|$ and $l(\mu) = l(\lambda) + 1$ by moving the $diag(\lambda)$ to the bottom of λ_{last} .

Now, if $diag(\lambda) = \lambda_{last}$ and they overlap, then $|\lambda| = m^2 + \binom{m}{2} = \frac{m(3m-1)}{2}$. If $diag(\lambda) = \lambda_{last} - 1$, then $|\lambda| = \frac{m(3m+1)}{2}$. $\square$

Definition 6.8. The *q-binomial coefficient*, denoted as $\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{[n]!_q}{[k]!_q [n-k]!_q}$.

Definition 6.9. $p(j, k, n) = \#\{\lambda \vdash n \mid \lambda_1 \leq j, l(\lambda) \leq k\}$ and $P(j, k) = \sum_{n=0}^{jk} p(j, k, n) q^n$.

Theorem 6.10. $P(j, k) = \begin{bmatrix} j+k \\ j \end{bmatrix}_q$.

Proof. If $l(\lambda) = k$, let $\mu = \lambda$ without the first column, then we have $\mu \vdash n - k$, $l(\mu) \leq k$, $\mu_1 \leq j - 1$.

If $l(\lambda) < k$, then we have $l(\lambda) \leq k - 1$.

Therefore, we have the recursive formula:

$$p(j, k, n) = p(j, k - 1, n) + p(j - 1, k, n - k).$$

Plugging the formula into the definition of $P(j, k)$, we have:

$$P(j, k) = P(j, k - 1) + P(j - 1, k)q^k.$$

Suppose we are inducting on $j + k$, then we have $P(j, k - 1) = \begin{bmatrix} j + k - 1 \\ j \end{bmatrix}_q$ and $P(j - 1, k) = \begin{bmatrix} j + k - 1 \\ k \end{bmatrix}_q$.

Plugging them into the above equality, we get:

$$\begin{aligned} P(j, k) &= P(j, k - 1) + P(j - 1, k)q^k \\ &= \begin{bmatrix} j + k - 1 \\ j \end{bmatrix}_q + \begin{bmatrix} j + k - 1 \\ k \end{bmatrix}_q q^k \\ &= \frac{[j + k - 1]_q!}{[j]_q! [k - 1]_q!} + \frac{[j + k - 1]_q!}{[j - 1]_q! [k]_q!} q^k \\ &= \frac{[j + k - 1]_q!}{[j]_q! [k]_q!} ([k]_q + [j]_q q^k) \\ &= \frac{[j + k - 1]_q!}{[j]_q! [k]_q!} \left(\frac{1 - q^k + q^k - q^{j+k}}{1 - q} \right) \\ &= \frac{[j + k - 1]_q!}{[j]_q! [k]_q!} [j + k]_q \\ &= \begin{bmatrix} j + k \\ j \end{bmatrix}_q \end{aligned}$$

as we desire. □

Theorem 6.11. $\prod_{j=0}^{n-1} (1 - q^j t) = \sum_{k=0}^n q^{\frac{k(k-1)}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q t^k.$

Proof. We have $\prod_{j=0}^{n-1} (1 - q^j t) = \sum_{k=0}^{n-1} \sum_{\substack{0 \leq j_1 < \dots < j_k \leq n-1}} q^{j_1 + \dots + j_k} t^k.$

Notice that $\{j_i\}$ can be uniquely mapped to $\{\lambda_i\}$ where $\lambda_i = j_i - i + 1$, giving us a unique partition λ , with $j_1 + \dots + j_k = \binom{k}{2} + |\lambda|$. Therefore, we have

$$\begin{aligned} \prod_{j=0}^{n-1} (1 - q^j t) &= \sum_k \sum_{\lambda} q^{|\lambda|} q^{\binom{k}{2}} t^k \\ &= \sum_k q^{\binom{k}{2}} t^k \sum_{\substack{\lambda, \lambda_1 \leq n-k, \\ l(\lambda) \leq k}} q^{|\lambda|} \\ &= \sum_k q^{\frac{k(k-1)}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q t^k. \end{aligned}$$

□

Theorem 6.12. $\prod_{j=0}^{n-1} \frac{1}{1 - q^j t} = \sum_{k=0} \left[\begin{matrix} n+k-1 \\ k \end{matrix} \right]_q t^k$

Proof.

$$\begin{aligned} \prod_{j=0}^{n-1} \frac{1}{1 - q^j t} &= \sum_{m_0, \dots, m_{n-1}} \prod_{j=0}^{n-1} (q^j t)^{m_j} \\ &= \sum_k \sum_{\substack{\lambda, l(\lambda) \leq n-1, \\ |\lambda| \leq k}} q^{|\lambda|} t^k \\ &= \sum_k P(n-1, k) t^k \\ &= \sum_k \left[\begin{matrix} n+k-1 \\ k \end{matrix} \right]_q t^k. \end{aligned}$$

□

Theorem 6.13. $\#\{V \subset \mathbb{F}_q^n, \dim(V) = k\} = \left[\begin{matrix} n \\ k \end{matrix} \right]_q$

Proof. Suppose V is generated by $v_1, \dots, v_k$. We have a $k \times n$ matrix with the i th row v_i . By row reduction we get some $k \times n$ matrix with k leading 1s and some non-zero elements on the $n - k$ non-pivot columns, marked by $*$'s, which uniquely determines the k -tuple that generates V . We observe that the $*$'s form a partition with $l(\lambda) \leq k$ and $\lambda_1 \leq n - k$. Since we have q choices for each $*$, and $q^{|\lambda|}$ choices for each λ , we get the number of $V = P(k, n - k) = \left[\begin{matrix} n \\ k \end{matrix} \right]_q$. □

Remark 6.14. One way for finding the number of full rank $k \times n$ matrices over $\mathbb{F}_q$ is to consider such matrices as a k -tuple of linearly independent vectors in $\mathbb{F}_q^n$.

We have $q^n - 1$ ways of deciding v_1 , $q^n - q$ ways of deciding v_2 , etc., since we have to make sure v_2 is linearly independent with v_1 (i.e. $v_2 = av_1$ for some $a \in \mathbb{F}_q$). Similarly, we have $q^n - q^2$ ways of deciding v_3 ($v_3 \neq av_1 + bv_2$ for any $a, b \in \mathbb{F}_q$, and any two sums in this form are different because v_1 and v_2 are linearly independent).

Therefore, we have in total

$$(q^n - 1)(q^n - q) \dots (q^n - q^{k-1})$$

different k -tuples of linearly independent vectors.

Similarly, we have

$$|GL_n(\mathbb{F}_q)| = (q^k - 1) \dots (q^k - q^{k-1}).$$

Notice that if $V = \mathbb{F}_q \langle v_1, \dots, v_k \rangle$ with $\dim V = k$, then we have the number of different $k \times n$ matrices over $\mathbb{F}_q$ generating V equal to the number of different invertible $k \times k$ matrices over $\mathbb{F}_q$ since we can always get a linear transformation by multiplying a $k \times k$ invertible matrix to A generating V . In other words, there are exactly $|GL_n(\mathbb{F}_q)|$ k -tuples of linearly independent vectors generate the same k -dimensional subspace V .

Therefore,

$$\begin{aligned}
 \#V &= \frac{(q^n - 1) \dots (q^n - q^{k-1})}{(q^k - 1) \dots (q^k - q^{k-1})} \\
 &= \frac{[n]_q \dots [n - k + 1]_q}{[k]_q \dots [1]_q} \\
 &= \frac{[n]_q!}{[k]_q! [n - k]_q!} \\
 &= \begin{bmatrix} n \\ k \end{bmatrix}_q
 \end{aligned}$$

as we desired.

Example 6.15. Let w be the arrangement of k 0s and $n - k$ 1s, and define $\text{inv}(w) = \#\{(i, j), i < j, w_i > w_j\}$.

Now, we consider w as a unique path from $(0, 0)$ to $(n - k, k)$ where a 0 means going up and an 1 means going right. We are able to get a unique partition λ with $l(\lambda) \leq k$ and $\lambda_1 \leq n - k$, and $|\lambda| = \text{inv}(w)$ since $\text{inv}(w)$ sums up the number of squares in each column.

Corollary 6.16.
$$\sum_{\substack{\lambda \vdash n, l(\lambda) \leq k \\ \lambda_1 \leq n - k}} q^{|\lambda|} = \sum_{\substack{w \text{ permutation of} \\ 0 \& 1}} q^{\text{inv}(w)}.$$

Remark 6.17. Let w be a permutation of k 0's and $n - k$ 1's. Let $\sigma \in S_k$ and $\pi \in S_{n-k}$. Now we replace the 0's by $\{1, 2, \dots, k\}$ in the order of σ and the 1's by $\{k + 1, \dots, n\}$ in the order of π . Now we have u a permutation of n elements and $\text{inv}(u) = \text{inv}(\sigma) + \text{inv}(\pi) + \text{inv}(w)$.

Thus, we have

$$\begin{aligned}
 \sum_{u \in S_n} q^{\text{inv}(u)} &= \sum_{w, \sigma, \pi} q^{\text{inv}(w) + \text{inv}(\pi) + \text{inv}(\sigma)} \\
 &= \sum_w q^{\text{inv}(w)} \sum_{\pi} q^{\text{inv}(\pi)} \sum_{\sigma} q^{\text{inv}(\sigma)}.
 \end{aligned}$$

Thus, we get $[n]_q! = [k]_q! [n - k]_q! \sum_{\substack{\lambda \vdash n, l(\lambda) \leq k, \\ \lambda_1 \leq n - k}} q^{|\lambda|}.$

Therefore,
$$\sum_{n=0}^{jk} P(j, k, n) q^n = \begin{bmatrix} n \\ k \end{bmatrix}_q.$$

7 Set Partitions

Notice that we have the following table about the number of functions $f : N \rightarrow X$ with $|N| = n$, $|X| = x$.

N	X	Any f	f injective	f surjective
$\{1, \dots, n\}$	$\{1, \dots, x\}$	x^n	$\prod_{i=0}^{n-1} (x - i)$	? $(x!S(n, x))$
Identical	$\{1, \dots, x\}$	$\binom{x}{n} = \binom{n+x-1}{x-1}$	$\binom{x}{n}$	$\binom{x}{n-x} = \binom{n-1}{x-1}$
$\{1, \dots, n\}$	Identical	? $(\sum S(n, k))$	1	? $(S(n, x))$
Identical	Identical	$p_{\leq x}(n)$	1	$p_x(n)$

To deal with the question marks, we have the following definition:

Definition 7.1. A **set partition** of $[n]$ into k parts is a set of subsets of S $\{B_1, \dots, B_k \mid B_i \cap B_j = \emptyset, \bigcup B_i = S, B_i \neq \emptyset\}$.

The **Stirling number of the second kind**, $S(n, k)$, denotes the number of set partitions of $[n]$ into k parts.

Example 7.2. $S(n, 1) = 1$, $S(n, 2) = (2^n - 2)/2 = 2^{n-1} - 1$, $S(n, n-1) = \binom{n}{2}$, $S(n, n) = 1$.

Theorem 7.3. $S(n, k) = S(n-1, k-1) + kS(n-1, k)$

Proof. We consider whether there is a j s.t. $B_j = \{n\}$.

If so, then removing B_j gives us a set partition of $n-1$ into $k-1$ parts ($S(n-1, k-1)$).

If not, then removing n from $B_j \ni n$ we get a set partition of $n-1$ into k parts, and we have k choices for where to put n ($kS(n-1, k)$).

Therefore, $S(n, k) = S(n-1, k-1) + kS(n-1, k)$. □

Definition 7.4. Let $F(n, k) = k!S(n, k)$, then $F(n, k)$ is the number of **ordered set partition** of $[n]$ into k parts. In other words, $F(n, k) = \#\{f : [n] \rightarrow [k], f \text{ surjective}\}$.

Corollary 7.5. $F(n, k) = kF(n-1, k-1) + kF(n-1, k)$

Proof. We consider whether there is a j s.t. $B_j = \{n\}$.

If so, then we remove B_j and we get an ordered set partition of $n-1$ into $k-1$ parts, and there are k choices for B_j , so the number is $kF(n-1, k-1)$.

If not, then removing n from $B_j \ni n$ we get an ordered set partition of $n-1$ into k parts, and we have k choices for where to put n ($kF(n-1, k)$).

Therefore, $F(n, k) = kF(n-1, k-1) + kF(n-1, k)$. □

Theorem 7.6. $\sum_{k=1}^n S(n, k)x(x-1)\dots(x-k+1) = x^n$

Proof. We have

$$\begin{aligned} \sum_{k=1}^n S(n, k)x(x-1)\dots(x-k+1) &= x^n = \sum_{k=1}^n k!S(n, k) \frac{x \dots (x-k+1)}{k!} \\ &= \sum_{k=1}^n F(n, k) \binom{x}{k} \end{aligned}$$

which represents the number of surjective functions from $[n]$ to all the subsets of $[x]$ in the form $\{x_1, \dots, x_k\}$, where $k \leq n$. This is exactly the number of functions f from $[n]$ to $[x]$, if we take $\{x_1, \dots, x_k\}$ to be the range of f . Therefore, the equality holds. □

Theorem 7.7. $\sum_{n=1}^{\infty} S(n, k) \frac{x^n}{n!} = \frac{(e^x - 1)^k}{k!}.$

Proof. Let $F_k(x) = \sum_{n=1}^{\infty} S(n, k) \frac{x^n}{n!} = \sum_{n=1}^{\infty} k S(n-1, k) \frac{x^n}{n!} + \sum_{n=1}^{\infty} S(n-1, k-1) \frac{x^n}{n!}.$

As a result, we have

$$\begin{aligned} \frac{d}{dx} F_k(x) &= \sum_{n=1}^{\infty} \frac{S(n-1, k-1)}{(n-1)!} x^{n-1} + \sum_{n=1}^{\infty} k \frac{S(n-1, k)}{(n-1)!} x^{n-1} \\ &= k F_k(x) + F_{k-1}(x) \end{aligned}$$

As a result, we get $F'_k - k F_k = F_{k-1}$, indicating that $(e^{-kx} F_k)' = e^{-kx} F_{k-1}.$

Solving the differential equation, we have that $F_k(x) = e^{kx} \int_0^x e^{-kt} F_{k-1}(t) dt + C \cdot e^{kx}.$

We prove that $F_k(x) = \frac{(e^x - 1)^k}{k!}$ by induction. Suppose that the equality holds for $k-1$, then we get

$$\begin{aligned} F_k(x) &= e^{kx} \int_0^x e^{-kt} F_{k-1}(t) dt + C \cdot e^{kx} \\ &= e^{kx} \int_0^x e^{-t} (e^{-t})^{k-1} \frac{(e^t - 1)^{k-1}}{(k-1)!} dt + C \cdot e^{kx} \\ &= e^{kx} \int_0^x \frac{(1 - e^{-t})^{k-1}}{(k-1)!} e^{-t} dt + C \cdot e^{kx} \\ &= e^{kx} \frac{(1 - e^{-t})^k}{k!} + C \cdot e^{kx} \Big|_0^x \\ &= e^{kx} \frac{(1 - e^{-x})^k}{k!} \\ &= \frac{(e^x - 1)^k}{k!} \end{aligned}$$

□

Theorem 7.8. $\sum_{n=1}^{\infty} S(n, k) x^n = \frac{x^k}{(1-x) \dots (1-kx)}$

Proof. Let $\sum_{n=1}^{\infty} S(n, k) x^n = G_k(x).$ We have

$$\begin{aligned} G_k(x) &= \sum k S(n-1, k) x^n + \sum S(n-1, k-1) x^n \\ &= kx G_k(x) + x G_{k-1}(x) \\ &= \frac{x}{1-kx} G_{k-1}(x) \\ &= \frac{x}{1-kx} \frac{x}{1-(k-1)x} G_{k-2}(x) \\ &= \dots = \frac{x^k}{(1-x) \dots (1-kx)} \end{aligned}$$

the last step holds since we have $G_1(x) = \sum x^n = \frac{x}{1-x}.$

□

Corollary 7.9. $S(n, k) = \sum_{\substack{\{n_1, \dots, n_k\} \subset \mathbb{N}^k, \\ \sum n_i = n-k}} 1^{n_1} 2^{n_2} \dots k^{n_k}$

Proof. $G_k(x) = \frac{x^k}{(1-x) \dots (1-kx)} = x^k (\sum x^n) (\sum 2^n x^n) \dots (\sum k^n x^n).$

$$\text{Therefore, } S(n, k) = \sum_{\substack{\{n_1, \dots, n_k\} \subset \mathbb{N}^k, \\ \sum n_i = n-k}} 1^{n_1} 2^{n_2} \dots k^{n_k}.$$

□

Remark 7.10. Notice that

$$\begin{aligned} F_k(x) &= \frac{(e^x - 1)^k}{k!} \\ &= \sum_{i=0}^k \frac{\binom{k}{i} e^{ix} (-1)^{k-i}}{k!} \\ &= \sum_i \sum_n \frac{\binom{k}{i} i^n \frac{x^n}{n!} (-1)^{k-i}}{k!} \end{aligned}$$

$$\text{Therefore, } S(n, k) = \sum_i \frac{\binom{k}{i} i^n (-1)^{k-i}}{k!}.$$

Multiplying by $k!$, we get

$$k!S(n, k) = \sum_{i=0}^n (-1)^{k-i} \binom{k}{i} i^n = k^n - \binom{k}{1} (k-1)^n + \dots + k(-1)^{k-1}.$$

8 Inclusion-Exclusion Principle

Theorem 8.1. Let $A_1, \dots, A_k$ be sets. Then we have

$$\begin{aligned} |A_1 \cup \dots \cup A_k| &= |A_1| + \dots + |A_k| - |A_1 \cap A_2| - \dots + (-1)^{k-1} |A_1 \cap \dots \cap A_k| \\ &= \sum_{r=1}^k \sum_{i_1 < \dots < i_r} (-1)^{r-1} |A_{i_1} \cap \dots \cap A_{i_r}| \end{aligned}$$

Proof. Let $A_{i_1}, \dots, A_{i_j}$ be the sets where x is in. Then x is counted exactly once because

$$\sum_{i=1}^j \binom{j}{i} (-1)^{i-1} = (1-1)^j + 1 = 1.$$

The formula therefore computes $|A_1 \cup \dots \cup A_k|$. □

Example 8.2. We have shown that $k!S(n, k) = \#\{f : [n] \rightarrow [k] \mid f \text{ surjective}\}$, which is equal to $k^n - \#\{f : [n] \rightarrow [k] \mid f \text{ not surjective}\}$.

Let $A_1 = \{f : [n] \rightarrow [k] \setminus \{1\}\}, \dots, A_k = \{f : [n] \rightarrow [k] \setminus \{k\}\}$, then

$$\begin{aligned} \#\{f : [n] \rightarrow [k] \mid f \text{ not surjective}\} &= |A_1 \cup \dots \cup A_k| \\ &= \sum_{i=1}^k \sum_{j_1 < \dots < j_i} (-1)^{i-1} |A_{j_1} \cap \dots \cap A_{j_i}| \\ &= \sum_{i=1}^k \sum_{j_1 < \dots < j_i} (-1)^{i-1} (k-i)^n \\ &= \sum_{i=1}^k (-1)^{i-1} \binom{k}{i} (k-i)^n \end{aligned}$$

Thus, the number of $\{f \text{ surjective}\}$ is $k^n - \sum_{i=1}^k (-1)^{i-1} \binom{k}{i} (k-i)^n = \sum_{i=0}^k (-1)^i \binom{k}{i} (k-i)^n$, which therefore equals $k!S(n, k)$.

Definition 8.3. A **derangement** is $w \in S_n$ with $w_i \neq i, \forall i$ (i.e. w has no cycle of length 1).

Theorem 8.4. $\frac{D(n)}{n!} = (1 - 1 + \frac{1}{2!} - \dots + \frac{(-1)^n}{n!})$ which approximates to e^{-1} as $n \rightarrow \infty$.

Proof. Let $D(n) = \#$ of derangements $= n! - \#\{w \in S_n, w_i = i \text{ for some } i \in [n]\}$.

Let $A_i = \{w \in S_n, w(i) = i\}$, then the number of derangements is $n! - |A_1 \cup \dots \cup A_n|$.

We also have

$$\begin{aligned} |A_1 \cup \dots \cup A_n| &= \sum_{k=1}^n (-1)^{k-1} \sum_{i_1 < \dots < i_k} |A_{i_1} \cap \dots \cap A_{i_k}| \\ &= \sum_{k=1}^n (-1)^{k-1} \binom{n}{k} (n-k)! \end{aligned}$$

Therefore, we have $|A_1 \cup \dots \cup A_n| = \sum_{k=1}^n (-1)^{k-1} \binom{n}{k} (n-k)! = \sum_{k=1}^n (-1)^{k-1} \frac{n!}{k!}$.

Therefore,

$$\begin{aligned}
 \text{the number of derangements} &= n! - n! \left(1 - \frac{1}{2!} + \frac{1}{3!} - \dots + (-1)^{n-1} \frac{1}{n!}\right) \\
 &= n! \left(1 - 1 + \frac{1}{2!} - \dots + (-1)^n \frac{1}{n!}\right) \\
 &\rightarrow n! \cdot e^{-1} \text{ as } n \rightarrow \infty.
 \end{aligned}$$

□

Corollary 8.5.

$$\begin{aligned}
 \sum_{n=0}^{\infty} D_n \frac{x^n}{n!} &= \sum_{n=0}^{\infty} \left(\sum_{k=0}^n (-1)^k \frac{1}{k!} \right) x^n \\
 &= \sum_{k=0}^{\infty} \sum_{n=k}^{\infty} (-1)^k \frac{x^n}{k!} \\
 &= \sum_{k=0}^{\infty} \frac{x^k (-1)^k}{k!} \sum_{n \geq k} x^{n-k} \\
 &= \frac{e^{-x}}{1-x}.
 \end{aligned}$$

Definition 8.6. Let S be an $[n]$ -set, V be the 2^n -dimensional vector space over R of all functions $f : 2^S \rightarrow R$. Let $\varphi : V \rightarrow V$ be the linear transformation defined by

$$\varphi f(T) = \sum_{Y \supset T} f(Y) \text{ for any } T \subset S.$$

In other words, let $M = \begin{bmatrix} 1 & \dots & 1 & \dots & 1 \\ \dots & \dots & \dots & \dots & \dots \\ 0 & \dots & 1 & \dots & 1 \\ \dots & \dots & \dots & \dots & \dots \\ 0 & \dots & 0 & \dots & 1 \end{bmatrix}$ where $M_{TY} = 1$ iff $T \subset Y$, then

$$M \begin{bmatrix} f(\emptyset) \\ \dots \\ f(T) \\ \dots \\ f(S) \end{bmatrix} = \begin{bmatrix} \varphi f(\emptyset) \\ \dots \\ \varphi f(T) \\ \dots \\ \varphi f(S) \end{bmatrix}.$$

Theorem 8.7. M^{-1} is defined by $M_{TY}^{-1} = (-1)^{|Y \setminus T|}$ if $Y \supset T$ and 0 otherwise.

In other words, $\varphi^{-1}f(T) = \sum_{Y \supset T} (-1)^{|Y \setminus T|} f(Y)$.

Proof. Let φ^{-1} be the linear transformation induced by M^{-1} . Then we have

$$\begin{aligned}
 \varphi\varphi^{-1}f(T) &= \sum_{Y \supset T} (-1)^{|Y \setminus T|} \varphi f(Y) \\
 &= \sum_{Y \supset T} (-1)^{|Y \setminus T|} \sum_{Z \supset Y} f(Z) \\
 &= \sum_{Z \supset T} \left(\sum_{Z \supset Y \supset T} (-1)^{|Y \setminus T|} \right) f(Z) \\
 &= \sum_{Z \supset T} f(Z) \sum_{i=0}^{|Z \setminus T|} (-1)^{|Z \setminus T|} \binom{|Z \setminus T|}{i} \\
 &= f(T)
 \end{aligned}$$

in which the last inequality holds because $\sum_{i=0}^{|Z \setminus T|} (-1)^{|Z \setminus T|} \binom{|Z \setminus T|}{i} = 1$ iff $Z = T$.

Therefore, M^{-1} is as claimed. $\square$

Example 8.8. Consider the number of k -compositions of n with no part equals to 2 ($a_1 + \dots + a_k = n, a_i \neq 2$).

We wish to find $\binom{n-1}{k-1} - \#\{k\text{-compositions of } n \text{ with } a_i = 2 \text{ for some } i\}$.

Let A_i be the set of k -compositions of n with $a_i = 2$, then

$$\begin{aligned}
 \binom{n-1}{k-1} - |A_1 \cup \dots \cup A_k| &= \binom{n-1}{k-1} - \sum_{i=1}^k (-1)^{i-1} \binom{k}{i} \binom{n-2i-1}{k-i-1} \\
 &= \sum_{i=0}^k (-1)^i \binom{k}{i} \binom{n-2i-1}{k-i-1}.
 \end{aligned}$$

To solve this by generating function, let $f_k(n)$ be the number, then we have

$$\begin{aligned}
 F_k(x) &= \sum_{n=0}^{\infty} f_k(n) x^n \\
 &= \sum_n \sum_{a_i \neq 2, \sum a_i = n} x^n \\
 &= \sum_n \sum_{a_i \neq 2, \sum a_i = n} x^{a_1 + \dots + a_k} \\
 &= \sum_{a_i \neq 2} x^{a_1 + \dots + a_k} \\
 &= \sum_{a_1 \neq 2} x^{a_1} \dots \sum_{a_k \neq 2} x^{a_k} \\
 &= (x + x^3 + x^4 + \dots)^k \\
 &= x^k (1 + x^2 + \dots)^k \\
 &= x^k \left(\frac{1}{1-x} - x \right)^k \\
 &= x^k \sum_{i=0}^k \binom{k}{i} \frac{(-x)^i}{(1-x)^{k-i}}.
 \end{aligned}$$

Definition 8.9. Let $S = \{j_1, \dots, j_k\}$. Then $d_n(S) = \#\{w \in S_n, D(w) = S\}$, $f_n(S) = \#\{w \in S_n, D(w) \subset S\}$.

Example 8.10. Let $n = 6$, $S = \{2, 5\}$, then $f_n(S)$ equals the set of $w \in S_6$ with $[6]$ splitted into 3 blocks and numbers in each one are in increasing order.

$$\text{Thus, } f_n(S) = \binom{6}{2} \binom{4}{3}.$$

In general,

$$\begin{aligned} f_n(S) &= \binom{n}{j_1} \binom{n-j_1}{j_2-j_1} \cdots \\ &= \frac{n!}{j_1!(j_2-j_1)!\cdots(n-j_k)!} \end{aligned}$$

Remark 8.11. Notice that $f_n(S) = \sum_{T \subset S} d_n(T)$. Consider the vector $(f_n(\emptyset), \dots, f_n([n-1]))^T$. It equals

$M \cdot (d_n(\emptyset), \dots, d_n([n-1]))^T$ where $M_{TY} = 1$ if $T \subset Y$.

Therefore, given $S = \{j_1, \dots, j_k\}$ we have

$$\begin{aligned} d_n(S) &= \sum_{T \subset S} (-1)^{|S \setminus T|} f_n(T) \\ &= \sum_{I \subset [k]} (-1)^{k-|I|} \frac{n!}{j_{i_1}!(j_{i_2}-j_{i_1})!\cdots} \\ &= n! \cdot \det C \end{aligned}$$

where C is a $k \times k$ matrix with $C_{rc} = \frac{1}{(j_c - j_{r-1})!}$ if $c \geq r-1$. Otherwise $C_{rc} = 0$. We manually define $j_0 = 0$ and $j_k = n$.

Example 8.12. Let $S = \{j_1, j_2\}$. We then have $d_n(S) = \frac{n!}{n!} - \frac{n!}{j_1!(n-j_1)!} - \frac{n!}{j_2!(n-j_2)!} + \frac{n!}{j_1!(j_2-j_1)!(n-j_2)!}$, corresponding to the case $T = \emptyset, \{j_1\}, \{j_2\}, S$, respectively.

$$\text{We also have } d_n(S) = n! \cdot \det \begin{bmatrix} \frac{1}{j_1!} & \frac{1}{j_2!} & \frac{1}{n!} \\ 1 & \frac{1}{(j_2-j_1)!} & \frac{1}{(n-j_1)!} \\ 0 & 1 & \frac{1}{(n-j_2)!} \end{bmatrix}, \text{ which is exactly the formula we gained above.}$$

9 The Rook Polynomial

Definition 9.1. For every permutation $w \in S_n$, we are able to get an $n \times n$ matrix A with $A_{ij} = 1$ iff $w_j = i$. Notice that such a matrix also represents one way for placing n non-attacking rooks on a $n \times n$ chessboard. Let $G(w)$ be $\{(i, j), A_{ij} = 1\}$.

Let $B \subset [n] \times [n]$ be a subset of squares on the chessboard.

Let $r_k(B) = \#$ ways to put k non-attacking rooks on B , $N_j = \#\{w \in S_n, |G(w) \cap B| = j\}$. In other words, N_j is the number of ways to put n non attacking rooks on the square with exactly j of them on B .

In particular, we have $r_1(B) = |B|$. If $B = \{(i, i), i \in [n]\}$, then we have $N_0 = \#\{w \in S_n, w(i) \neq i\} = D_n$.

$$\text{Let } N_B(x) = \sum_{j=0}^n N_j x^j.$$

Theorem 9.2. $N_B(x) = \sum_{k=0}^n r_k(B)(n-k)!(x-1)^k$.

Proof. Let $y = x - 1$. We then need to show $N_B(y+1) = \sum_{k=0}^n r_k(B)(n-k)!(y)^k$.

We have

$$\begin{aligned} N_B(y+1) &= \sum_j N_j (y+1)^j \\ &= \sum_j N_j \sum_k \binom{j}{k} y^k \\ &= \sum_k y^k \sum_{j=k}^n N_j \binom{j}{k}. \end{aligned}$$

Thus, it suffices to show that $\sum_{j=k}^n N_j \binom{j}{k} = r_k(B)(n-k)!$.

Notice that the LHS stands for having j non-attacking rooks on B and color k of them green. On the other hand, the RHS stands for having k non-attacking rooks on B and $n-k$ non-attacking rooks on the rest of G . Combinatorially we can see that the two sides are the same. $\square$

Corollary 9.3. $N_0 = N_B(0) = \sum_{k=0}^n r_k(B)(n-k)!(-1)^k$.

Example 9.4. Let $M_n = \#\{w \in S_n, w_i \neq i, i+1 \pmod m\}$. Let $B = \{(i, i), (i, i+1 \pmod m), i \in [n]\}$.

Then we have $M_n = N_0(B) = \sum_k r_k(B)(n-k)!(-1)^k$. Consider $[2n]$ put in a cycle, then $r_k(B) = \#$ ways to choose k non-consecutive points on $[2n]$.

If 1 is chosen, then we have $3 \leq x_1 < \dots < x_{k-1} \leq 2n-1$ with $x_{i+1} > x_i + 1$.

Equivalently, we have $3 \leq x_1 < x_2 - 1 < \dots < x_{k-1} - (k-2) \leq 2n-k+1$, so we have in total $\binom{2n-k+1-2}{k-1} = \binom{2n-k-1}{k-1}$ ways of choosing the rest points.

On the other hand, suppose 1 is not chosen, then we have $2 \leq x_1 < \dots < x_k \leq 2n$ with $x_{i+1} > x_i + 1$.

Equivalently, we have $2 \leq x_1 < x_2 - 1 < \dots < x_k - (k-1) \leq 2n-k+1$, so we have in total $\binom{2n-k}{k}$ ways of choosing the k points.

Therefore, $r_k(B) = \binom{2n-k}{k} + \binom{2n-k-1}{k-1}$. Plugging this back to this initial formula leads us to a formula of M_n .

Theorem 9.5. *If $B \subset [n] \times [n]$ is a Young diagram (e.g. $\lambda = (4, 3, 3, 2)$), $l(\lambda) = m$, then the maximum number of non-attacking rooks to put on B is exactly m .*

Moreover, we then have

$$\sum_{k=0}^m r_k(B) \cdot (x)_{m-k} = \prod_{i=1}^m (x + \lambda_i - m + i), \text{ where } \lambda_i \text{ is the } i\text{th index of } \lambda,$$

where $(x)_{m-k} = x(x-1)\dots(x-(m-k)+1)$.

Proof. Let x be a natural number. We add a $m \times x$ rectangle to the left of B . LHS represents the number of ways to put m non-attacking rooks on the union of B and the rectangle, since suppose that there are k rooks on B , then we can put $m-k$ rooks on the rectangle in $(x)_{m-k}$ ways.

On the other hand, RHS represents the number of ways to put m rooks on the union of B and the rectangle too, if we start from bottom and choose where to put rook on each row.

Therefore, we have the equality. $\square$

Remark 9.6. *If $\lambda = (m, m-1, \dots, 1)$, then $\sum_{k=0}^m r_k(B)(x)_{m-k} = x^m$, and we have $r_k(B) =$ the Stirling number $S(m, m-k)$ by a bijection.*

The bijection is as follows: suppose a rook is at (i, j) , then we say i and j are in the same subset.

The procedure offers a set partition of $[m]$ into $[m-k]$ subsets.

Definition 9.7. *Given a grid. Fix $A_1, \dots, A_n, B_1, \dots, B_n$ with A_i on the top-left of A_{i+1} and B_i on the top-left of B_{i+1} . A **set of non-intersecting lattice paths** is a set of lattice paths $\{\gamma_i : A_i \rightarrow B_i\}$ (only going up or right) with all the paths disjoint.*

Theorem 9.8. $\#NILP(A_1 \rightarrow B_1, \dots, A_n \rightarrow B_n) = \det [\#paths, A_i \rightarrow B_j]_{i,j=1}^k$.

For instance, let $A_1 = (0, 1), A_2 = (1, 0), A_3 = (3, 0), B_1 = (3, 5), B_2 = (4, 5), B_3 = (6, 4)$, then $\#NILP(A_1 \rightarrow B_1, A_2 \rightarrow B_2, A_3 \rightarrow B_3)$ is

$$\det \begin{bmatrix} \binom{7}{3} & \binom{8}{4} & \binom{9}{3} \\ \binom{7}{7} & \binom{8}{8} & \binom{9}{9} \\ \binom{2}{5} & \binom{3}{6} & \binom{4}{7} \\ \binom{0}{0} & \binom{1}{1} & \binom{3}{3} \end{bmatrix}$$

Proof. $\det [\#paths, A_i \rightarrow B_j]_{i,j=1}^k = \sum_{\sigma \in S_k} \text{sgn}(\sigma) \#A_1 \rightarrow B_{\sigma(1)} \dots \#A_k \rightarrow B_{\sigma(k)}.$

Notice that we have a bijection between $(A_1 \rightarrow B_{\sigma(1)}, \dots, A_k \rightarrow B_{\sigma(k)})$ and $(A_1 \rightarrow B_{\pi(1)}, \dots, A_k \rightarrow B_{\pi(k)})$ if $\pi = \sigma(i, j)$. Therefore, if we look for the minimum i s.t. $A_i \rightarrow B_{\sigma(i)}$ intersects with $A_j \rightarrow B_{\sigma(j)}$ at a point X , then X is also the intersection of $A_i \rightarrow B_{\sigma(j)}$ and $A_j \rightarrow B_{\sigma(i)}$.

Therefore, $\det [\#paths, A_i \rightarrow B_j]_{i,j=1}^k$ is the number of all non-intersecting paths from $A_i \rightarrow B_{\sigma(i)}$, and the only such paths are from $A_1 \rightarrow B_1, \dots, A_k \rightarrow B_k$.

Therefore, $\#NILP(A_1 \rightarrow B_1, \dots, A_n \rightarrow B_n) = \det [\#paths, A_i \rightarrow B_j]_{i,j=1}^k$. $\square$

10 The Catalan Number

Definition 10.1. The *Catalan number* C_n is the number of paths from $(0,0)$ to (n,n) not crossing below the main diagonal. Such a path is called a *Dyck path*.

Theorem 10.2. $C_n = \frac{1}{n+1} \binom{2n}{n}$.

1st Proof. To compute C_n , notice that if a path from $(0,0)$ to (n,n) crosses below the main diagonal, then it intersects the $(1,0) - (n,n-1)$ diagonal somewhere.

Let's consider a path from $(0,0)$ to $(n+1,n-1)$, and X be the first time the path intersects with the $(1,0) - (n,n-1)$ diagonal (which apparently has to exist). We then reflect the path from X to $(n+1,n-1)$ by the $(1,0) - (n,n-1)$ diagonal and get a path from $(0,0)$ to (n,n) (which crosses below the main diagonal). The corresponding paths form a bijection and we thus have $\binom{2n}{n} - C_n = \binom{2n}{n-1}$ so

$$C_n = \frac{1}{n+1} \binom{2n}{n}. \quad \square$$

2nd Proof. Let (k,k) be the first time the path intersects the main diagonal. Then we know our path (before (k,k)) is a Dyck path from $(0,1)$ to $(k-1,k)$, the number of which is C_{k-1} . On the other hand, our path from (k,k) is a Dyck path from (k,k) to (n,n) , the number of which is C_{n-k} .

$$\text{Therefore, } C_n = \sum_k C_{n-k} C_{k-1}.$$

Now we use generating function to show the result. We have

$$\begin{aligned} C(x) &= C_0 + \sum_{n=1}^{\infty} C_n x^n \\ &= C_0 + \sum_{n=1}^{\infty} \sum_{k=1}^n C_{k-1} C_{n-k} x^n \\ &= C_0 + \sum_{m=0}^{\infty} \sum_{k=1}^{\infty} C_{k-1} C_m x^{m+k} \\ &= C_0 + \sum_{k=1}^{\infty} C_{k-1} x^k \sum_{m=0}^{\infty} C_m x^m \\ &= C_0 + x C(x)^2. \end{aligned}$$

Therefore, $C(x) = 1 + xC(x)^2 = \frac{1 \pm \sqrt{1-4x}}{2x}$. Since $C(x) \rightarrow 1$ when $x \rightarrow 0$, we have

$$\begin{aligned} C(x) &= \frac{1 - \sqrt{1-4x}}{2x} \\ &= \frac{1}{2x} \left(1 - 1 - \sum_{k=1}^{\infty} \binom{1/2}{k} (-4x)^k \right) \\ &= \frac{1}{2x} \left(- \sum_{k=1}^{\infty} \frac{(1/2) \dots (1/2 - k + 1)}{k!} (-4)^k x^k \right) \\ &= \frac{1}{2x} \left(\sum_{k=1}^{\infty} \frac{1 \cdot 1 \cdot 3 \cdot \dots \cdot (2k-3)}{2^k k!} 4^k x^k \right) \\ &= \frac{1}{2x} \left(\sum_{k=1}^{\infty} \frac{1 \cdot 1 \cdot 3 \cdot \dots \cdot (2k-3) 2^k (k-1)!}{k! (k-1)!} x^k \right) \\ &= \frac{1}{2x} \left(\sum_{k=1}^{\infty} \frac{2(2k-2)!}{k! (k-1)!} x^k \right) \\ &= \sum_{k=1}^{\infty} \frac{1}{k} \binom{2k-2}{k-1} x^{k-1}. \end{aligned}$$

We thus have $C_n = \frac{1}{n+1} \binom{2n}{n}$. □

Theorem 10.3. # rooted plane binary tree with n nodes is C_n .

Theorem 10.4. # triangulations of an $n+2$ -gon is C_n .

Definition 10.5. A permutation on n elements **avoids** $\sigma \in S_k$ if $\nexists i_1, \dots, i_k$ s.t. $(w_{i_1}, \dots, w_{i_k}) \sim \sigma$ (i.e. $w_{i_j} < w_{i_r} \Leftrightarrow \sigma_j < \sigma_r$).

Theorem 10.6. Let $w \in S_n$, $\sigma = (132)$, then the number of w avoiding σ is C_n .

Proof. Let $w_k = n$, then $w = (w_1 \dots w_{k-1})n(w_{k+1} \dots w_n)$ and the first $k-1$ indices form a permutation of $k-1$ elements $(n-k+1, \dots, n-1)$ avoiding σ and the last $n-k$ indices form a permutation of $n-k$ elements $(1, \dots, n-k)$ avoiding σ .

We thus have $\#_n = \sum \#_{k-1} \cdot \#_{n-k}$ which, by induction, shows that $\#_n = C_n$. □

Theorem 10.7. If $\sigma \in S_3$, then the number of $w \in S_n$ avoiding σ is C_n .

11 Partially Ordered Set

Definition 11.1. A *partial order* $\prec$ on a set X satisfies

- (i) $x \preceq x$.
- (ii) If $x \preceq y$, $y \preceq x$, then $x = y$.
- (iii) If $x \preceq y$, $y \preceq z$, then $x \preceq z$.

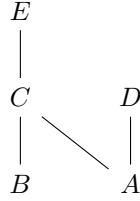
A **covering relation** is denoted $x \prec y$, which happens iff $x \prec y$ and $\nexists z$ s.t. $x \prec z \prec y$.
 $x \parallel y$ if they are not comparable.

Definition 11.2. A **poset (partially ordered set)** $P = (X, \prec)$ is a set of elements X together with a partial order $\prec$ defined on the set X .

Definition 11.3. The **Hasse Diagram** is the graph representation of a poset. It is formed by drawing the covering relations.

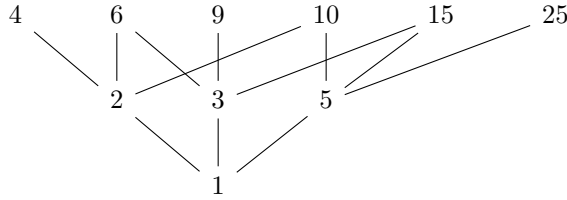
Example 11.4. (i) Let $\prec$ be defined on $\mathbb{R}^2$ s.t. $(x, y) \preceq (z, t)$ iff $x \leq z$ and $y \leq t$.

Let $X = \{A = (2, 1), B = (1, 2), C = (3, 3), D = (5, 1), E = (4, 4)\}$, then the Hasse diagram is as follows



(ii) Let the n -th chain C_n be the poset defined on $[n]$ with $\prec = <$, then the Hasse diagram is a straight line from 1 on the bottom to n on the top.

(iii) Let $X = \mathbb{N}$, $x \prec y$ if $x \mid y$, then (a part of) the Hasse diagram is as follows:



(iv) Let $X = \{S \subset [n]\}$ with $S \prec T$ iff $S \subset T$. Then the poset is named the **Boolean poset** and is isomorphic to $\{0, 1\}^n$.

Definition 11.5. Two posets are **isomorphic** if $X_P \cong X_Q$ by a map f and $s \prec_P t \Leftrightarrow f(s) \prec_Q f(t)$.

Definition 11.6. The **Young Lattice** $\mathbb{Y}$ is defined as $X = \{\lambda \text{ integer partition}\}$ and $\mu \prec \lambda$ iff $\mu \subset \lambda$ ($\mu_i < \lambda_i, \forall i$).

Definition 11.7. Given a poset and its Hasse diagram, we can form a total order by assigning each vertex a positive integer from 1 to $|P|$ that respects the order (without repetitive use of the integers). Such a labelling is called a **linear extension**.

The number of linear extension of P is denoted as $e(P)$.

Example 11.8. $e(C_n) = 1$. $e(C_n \sqcup C_m) = \binom{m+n}{m}$.

Conjecture 11.9. For every finite poset $P \neq C_n$, there exists vertices x and y s.t. $\frac{e(P, x < y)}{e(P)} \in [\frac{1}{3}, \frac{2}{3}]$.

Definition 11.10. A *chain* in a poset P is a subset $\{x_1, \dots, x_n\}$ s.t. $x_1 \preceq x_2 \preceq \dots \preceq x_n$.

A *antichain* in a poset P is a subset $\{x_1, \dots, x_n\}$ s.t. $x_i \parallel x_j, \forall i, j$.

Corollary 11.11. The number of maximal chains in a Boolean poset on $[n]$ is $n!$.

Definition 11.12. In $\mathbb{Y}$, we label the Young diagrams by $[n]$ s.t. $T(i, j) < T(i+1, j), T(i, j) < T(i, j+1)$. The resulted table is called a **Young Tableaux**.

Corollary 11.13. Let $X = \{(i, j)\}$ representing λ and let $(i, j) \prec (i, j+1), (i, j) \prec (i+1, j)$. Then X is given a partial order and becomes a poset.

The number of linear extensions of this poset is in bijection with the number of different Young tableau on the Young diagram of λ .

12 Ranked Posets

Definition 12.1. A poset P is **ranked** if $\exists f : X \rightarrow \mathbb{N}$ s.t.

- (i) $x \prec y \Rightarrow f(x) < f(y)$.
- (ii) $x \prec y \Rightarrow f(y) = f(x) + 1$.
- (iii) If x is the minimal element of X (if it exists) then $f(x) = 0$.

Definition 12.2. A chain $\{x_1, \dots, x_n\}$ is called **maximal** if $\nexists y \in P$ s.t. $\{x_1, \dots, x_n, y\}$ is also a chain.

In other words, we have $x_1 \prec x_2 \prec \dots \prec x_n$.

Definition 12.3. A poset P is **graded** if every maximal chain has the same length, denoted as $h(P)$.

Corollary 12.4. Let P be graded and the rank function be $\rho : P \rightarrow \mathbb{N}$. Then $\rho(x) = 0 \Leftrightarrow x$ minimal; $\rho(x) = h(P) \Leftrightarrow x$ maximal.

P is ranked is equivalent to P being graded.

Proof. Suppose P is ranked. Let $x_1 \prec \dots \prec x_k$ be a maximal chain. We then have $\rho(x_k) = k - 1$, so every maximal chain has the same length and P is graded.

Now suppose P is graded. For every maximal chain $C : x_1 \prec \dots \prec x_k$, let $\rho(x_i) = i - 1$.

Suppose $\exists x = x_i = y_j$ in two maximal chains C and C' s.t. $i \neq j$. Suppose $i < j$. P is graded so C and C' are of the same length. Therefore, the chain $y_1 \prec \dots \prec y_j \prec x_{i+1} \prec \dots \prec x_k$ is also a maximal chain and has length greater than $k - 1$, which leads to contradiction.

Therefore, ρ is well-defined and P is ranked. $\square$

Definition 12.5. The **rank generating function** is defined as

$$F(P, x) = \sum_{i=0}^{n-1} p_i x^i,$$

where p_i is the number of elements in P with rank i .

Definition 12.6. The poset on the set partitions of $[n]$ is denoted as Π_n .

The partial order is defined as $\{X_1, \dots, X_k\} \prec \{X_1, \dots, X_i \cup X_j, \dots, X_k\}$. We say π is a **refinement** of σ if $\pi \preceq \sigma$.

Corollary 12.7. (i) $F([n], x) = 1 + x + \dots + x^{n-1}$.

(ii) $F(B_n, x) = (1 + x)^n$.

(iii) $F(\Pi_n, x) = \sum_{k=0}^{n-1} S(n, n-k) x^k$.

Definition 12.8. $[s, t] = \{p \in P, s \preceq p \preceq t\}$.

The **order ideal** of P is a set $I \subset P$ s.t. if $x \in I$ and $y \preceq x$ then $y \in I$.

Definition 12.9. The **dual poset** P^* of a poset P is defined as $(X, \prec^*)$ s.t. $s \prec^* t$ iff $t \prec s$ in P .

$P + Q = (X_P \sqcup X_Q, \prec)$ s.t. $s \prec t$ iff $s, t \in P$ and $s \prec_P t$ or $s, t \in Q$ and $s \prec_Q t$.

$P \oplus Q = (X_P \sqcup X_Q, \prec)$ s.t. $s \prec t$ iff $s, t \in P$ and $s \prec_P t$ or $s, t \in Q$ and $s \prec_Q t$ or $s \in P, t \in Q$.

$P \times Q = (X_P \times X_Q, \prec_P \times \prec_Q)$.

Corollary 12.10. (i) $(P \oplus Q) \oplus R = P \oplus (Q \oplus R)$.

(ii) $[1] \oplus [1] \oplus \dots \oplus [1] = [n]$.

(iii) $[2] \times [2] \times \dots \times [2] \cong B_n$.

(iv) $[n] \times [m] \cong n \times m$ grid.

(v) P graded $\Leftrightarrow P^*$ graded.

- (vi) $P + Q$ graded $\Leftrightarrow P, Q$ graded and $h(P) = h(Q)$.
- (vii) $P \oplus Q$ graded $\Leftrightarrow P, Q$ graded.
- (viii) $F(P \oplus Q, x) = F(P, x) + x^n F(Q, x)$ where $n = h(P)$.
- (ix) $\rho_{P \times Q}(s, t) = \rho_P(s) + \rho_Q(t)$.

13 Lattice

Definition 13.1. A poset P is a lattice if $\forall s \neq t \in P, \exists!$ minimal u s.t. $s \leq u, t \leq u$ and $\exists!$ maximal v s.t. $v \leq s, v \leq t$.

The minimal u is called the **join** of s and t , denoted $s \vee t$.

The maximal v is called the **meet** of s and t , denoted $s \wedge t$.

Example 13.2. (i) Chain C_n is a lattice. $\forall i \leq j, i \vee j = j$ and $i \wedge j = i$.

(ii) Antichain A_n is not a lattice as there is no comparison.

(iii) A $m \times n$ grid is a lattice with $(x, y) \vee (z, w) = (\max(x, z), \max(y, w))$, $(x, y) \wedge (z, w) = (\min(x, z), \min(y, w))$.

(iv) B_n is a lattice with $s \vee t = s \cup t$ and $s \wedge t = s \cap t$.

(v) $\mathbb{Y}$ is a lattice with $\mu \vee \lambda = \mu \cup \lambda$ and $\mu \wedge \lambda = \mu \cap \lambda$.

Theorem 13.3. If P is a finite lattice, then $\exists!$ minimal element and maximal element, denoted as $\hat{0}$ and $\hat{1}$.

Proof. Let $\hat{0} = \bigwedge v_i$, then we know that $\hat{0}$ is the minimal element. It is unique because if $0'$ is another minimal element then $\hat{0} \wedge 0' = \hat{0} = 0'$.

The proof for the existence and uniqueness of the maximal element is analogous. $\square$

Proposition 13.4. Let L be a finite lattice, then

(i) L graded with rank function ρ s.t. $\rho(s) + \rho(t) \geq \rho(s \vee t) + \rho(s \wedge t)$.

(ii) If $s \wedge t \prec s, t$, then $s, t \prec s \vee t$.

are equivalent.

Proof. Suppose i) holds and $s \wedge t \prec s, t$. Then we have $\rho(s \wedge t) = \rho(s) - 1 = \rho(t) - 1$. And $\rho(s \vee t) = \rho(s) + 1 = \rho(t) + 1$. Thus, $s, t \prec s \vee t$.

On the other hand, suppose ii) holds. Let $[u, v]$ be an interval of L with minimal length and not graded. Then there are s_1, s_2 covering u s.t. $[s_1, v], [s_2, v]$ are graded with length l_1, l_2 , and $l_1 \neq l_2$.

By ii), we have saturated chains $s_i \prec s_1 \vee s_2 \prec t_1 \prec \dots \prec t_k = v$, indicating that $l_1 = l_2$, so L is graded.

Now, suppose that there exist $s, t \in L$ satisfying that

$$\rho(s) + \rho(t) < \rho(s \vee t) + \rho(s \wedge t)$$

with $s \wedge t, s \vee t$ minimal and then $\rho(s) + \rho(t)$ minimal.

We therefore have s or t not covering $s \wedge t$; suppose it is s . There then exists s' s.t. $s \wedge t \prec s' \prec s$. Notice that we then have

$$\rho(s') + \rho(t) \geq \rho(s' \vee t) + \rho(s' \wedge t)$$

since $l(s' \wedge t, s' \vee t) < l(s \wedge t, s \vee t)$ but $l(s \wedge t, s \vee t)$ and $\rho(s) + \rho(t)$ are minimal.

As we have $s' \wedge t = s \wedge t$, we then have

$$\rho(s) + \rho(s' \vee t) < \rho(s') + \rho(s \vee t).$$

As we have $s' \preceq s \wedge (s' \vee t)$ and $s \vee t = s \vee (s' \vee t)$, we then have $S = s$ and $T = s' \vee T$ satisfying that $\rho(S) + \rho(T) < \rho(S \vee T) + \rho(S \wedge T)$, which contradicts the claim as $l(S \wedge T, S \vee T) < l(s \wedge t, s \vee t)$. $\square$

Definition 13.5. A **distributive lattice** is a lattice D s.t.

$$s \vee (t \wedge u) = s \vee t \wedge s \vee u.$$

$$s \wedge (t \vee u) = s \wedge t \vee s \wedge u.$$

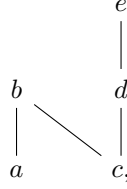
for any $s, t, u \in P$.

Example 13.6. B_n is a distributive lattice.

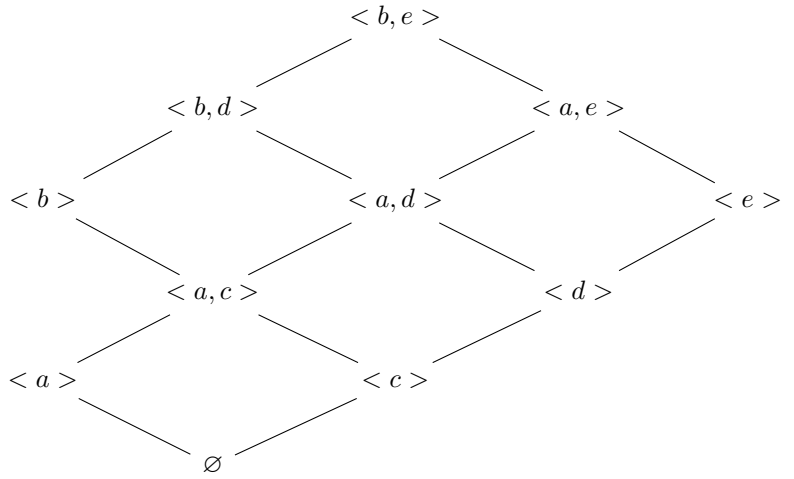
Theorem 13.7 (Fundamental Theorem of Finite Distributive Lattice). L is finite distributive lattice iff $L \cong J(P)$ for some poset P , where $J(P)$ is a lattice of order ideals of P .

In $J(P)$, $I_1 \prec I_2 \Leftrightarrow I_1 \subset I_2$.

Example 13.8. Let P be the following poset:



then $J(P)$ looks like the following:



which is a distributive lattice by set operation laws, since we have $I_1 \wedge I_2 = I_1 \cap I_2$, $I_1 \vee I_2 = I_1 \cup I_2$.

Corollary 13.9. $B_n \cong J(\{a_1, \dots, a_n\})$, where $a_i \parallel a_j$, for any $i \neq j$.

Remark 13.10. Given L distributive lattice. Let $X \subset L$ s.t. $l \in X$ iff $\exists! u \prec l$. Then $P \cong L|_X$ gives $L \cong J(P)$.

Example 13.11. $J(C_m \cup C_n) \cong$ a $m \times n$ grid.

A lattice path from $\emptyset$ to $C_m \cup C_n$ in $J(C_m \cup C_n)$ is exactly a linear extension of $C_m \cup C_n$.

The number of such linear extensions is thus equal to the number of lattice paths, which is $\binom{m+n}{n}$.

Theorem 13.12. The following three sets are in bijection with each other in a poset P :

- (i) Order preserving maps $f : P \rightarrow [m]$ (i.e., $s \prec t \Rightarrow f(s) < f(t)$).
- (ii) Chains $\emptyset = I_0 \preceq I_1 \preceq \dots \preceq I_m = P$ in $J(P)$.
- (iii) Elements in $J(P \times C_{m-1})$.

Proof. Given $f : P \rightarrow [m]$, by letting $I_j = I_{j-1} \cup f^{-1}(j)$, we get a chain $\emptyset = I_0 \preceq I_1 \preceq \dots \preceq I_m = P$ in $J(P)$.

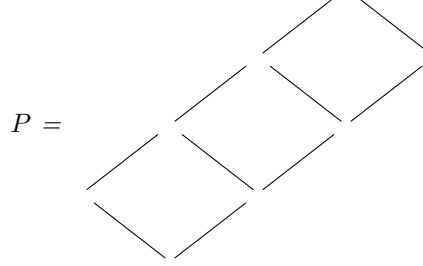
Similarly, given such a chain, let $f(x) = j$ iff $x \in I_j \setminus I_{j-1}$, we get an order preserving map $P \rightarrow [m]$.

Therefore, the first two sets are in bijection with each other.

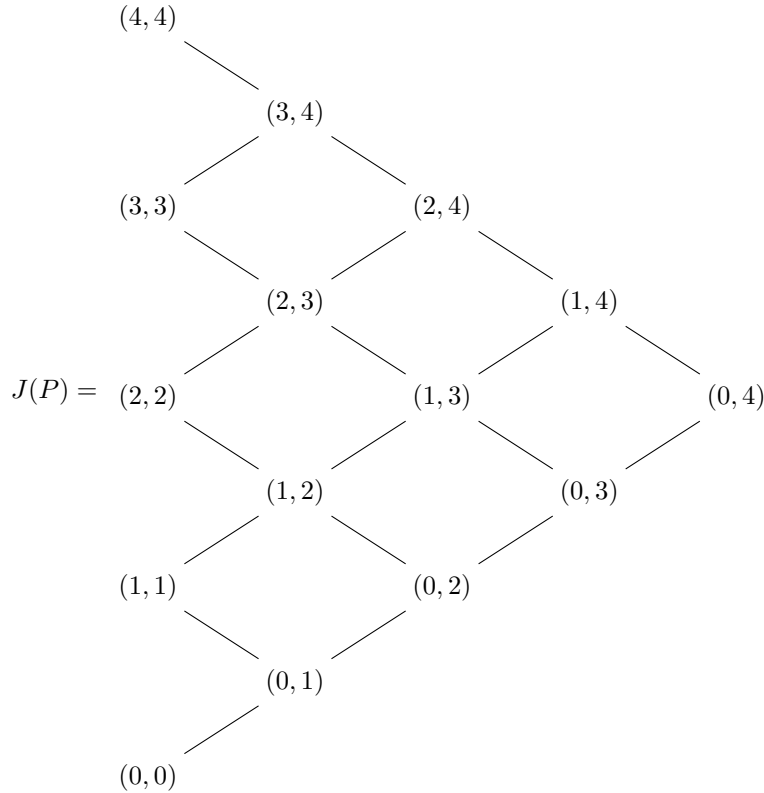
Now, suppose we are given a chain $\emptyset = I_0 \preceq I_1 \preceq \dots \preceq I_m = P$ in $J(P)$. Define $J \in J(P \times C_{m-1})$ to be the set $\{(t, j), t \in I_{m-j}\}$. J is indeed an order ideal, as if $s \prec t$ then $s \in I_{m-j}$ so $(s, j) \in J$, and if $k < j$ then $t \in I_{m-j} \subset I_{m-k}$, so $(t, k) \in J$.

On the other hand, suppose $J \in J(P \times C_{m-1})$ is given, define $f(t) = \min\{(m-j) \mid (t,j) \in J\}$ for each t . This indeed defines an order preserving function, as if $(t,j) \in J$ then $(s,j) \in J$ for any $s \prec j$. Therefore, $f(s) \leq f(t)$. $\square$

Example 13.13. The following grid P has $J(P)$ as follows:



and



where $(i,j) \in J(P)$ implies we are at the i -th point on the upper line and j -th point on the lower line.

Notice that $e(P) = \#$ linear extensions of P , which also equals the number of lattice paths in $J(P)$ (i.e., $\emptyset = I_0 \prec \dots \prec I_n = P$ with $|P| = n$). Therefore, $e(P) = C_n$ the Catalan number in this case.

Corollary 13.14. $e(P) = \#\{\emptyset = I_0 \prec \dots \prec I_n = P\}$, where $n = |P|$.

$e(P) = \sum_{s \in M} e(P \setminus \{s\})$ where M is the set of maximal elements in P .

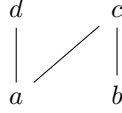
14 Incidence Algebra

Definition 14.1. Consider the functions $f : \{(s, t) \in P \times P \mid s \preceq t\} \rightarrow \mathbb{C}$ (here $\mathbb{C}$ can be replaced by an arbitrary field), where P is a finite poset.

All such functions form an algebra, denoted as $I(P)$, called the **incidence algebra** of P .

As a vector space, $I(P)$ is a subspace $I_P \subset \mathbb{C}^{\# \{s \preceq t\}}$. As a ring, $(f + g)(s, t) = f(s, t) + g(s, t)$ and $(f \cdot g)(s, t) = \sum_{s \preceq u \preceq t} f(s, u) \cdot g(u, t)$ are the addition and multiplication operations in $I(P)$.

Example 14.2. Let P be the poset drawn below



, we then have

$$I(P) = \{(f(a, a), f(a, c), f(a, d), f(b, b), f(b, c), f(c, c), f(d, d))\} \subset \mathbb{C}^7.$$

Definition 14.3. Finding a linear extension of P (which always exists) and ordering the elements in P based on it, we are then able to define a matrix $M(f)$ for each function $f \in I(P)$, where

$$M(f)_{ij} = \begin{cases} f(v_i, v_j) & v_i \preceq v_j \\ 0 & \text{otherwise} \end{cases}$$

Example 14.4. In the above example, we have

$$M(f) = \begin{bmatrix} f(a, a) & 0 & f(a, c) & f(a, d) \\ 0 & f(b, b) & f(b, c) & 0 \\ 0 & 0 & f(c, c) & 0 \\ 0 & 0 & 0 & f(d, d) \end{bmatrix}.$$

It is always true that $M(f)$ is upper triangular for any f .

Corollary 14.5. Thus, $f \in I(P) \iff$ upper triangular matrix $M(f) \in C^{n \times n}$ (s.t. $M(f)_{ij} = 0$ if $i \parallel j$ or $j \prec i$) is a bijection.

Corollary 14.6. $M(f) + M(g) = M(f + g)$.

$M(f) \cdot M(g) = M(f \cdot g)$.

Proof. The addition part is trivial.

For the multiplication part, notice that

$$\begin{aligned} [M(f)M(g)]_{ij} &= \sum_{k=1}^n M(f)_{ik} M(g)_{kj} \\ &= \sum_{v_i \preceq v_k \preceq v_j} f(v_i, v_k) g(v_k, v_j) \\ &= (f \cdot g)(i, j). \end{aligned}$$

□

Definition 14.7. The identity in $I(P)$ is denoted as δ , where $\delta(s, t) = 1$ iff $s = t$ and otherwise $\delta(s, t) = 0$. $M(\delta) = I$.

The **Zeta function** $\zeta(s, t)$ is defined as

$$\zeta(s, t) = \begin{cases} 1 & s \preceq t \\ 0 & \text{otherwise} \end{cases}.$$

Proposition 14.8. $\zeta^m(s, t) = \#\{(w_1, \dots, w_{m-1}) : s \preceq w_1 \preceq \dots \preceq w_{m-1} \preceq t\}$.

Proof.

$$\begin{aligned} \zeta^m(s, t) &= [M(\zeta)^m]_{st} \\ &= \zeta \cdot \zeta^{m-1}(s, t) \\ &= \sum_{s \preceq u_1 \preceq t} \zeta^{m-1}(u, t) \\ &= \dots = \sum_{s \preceq u_1 \preceq \dots \preceq u_{m-1} \preceq t} 1, \end{aligned}$$

which is exactly what the RHS of the proposition. $\square$

Corollary 14.9. We have

$$M(\zeta - \delta) = \begin{bmatrix} 0 & * & \dots & * \\ 0 & 0 & \dots & * \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & 0 \end{bmatrix}$$

which is always nilpotent.

Corollary 14.10. $(\zeta - \delta)^m(s, t) = \#\{(w_1, \dots, w_{m-1}) : s \prec w_1 \prec \dots \prec w_{m-1} \prec t\}$

Proof. We have

$$\begin{aligned} (\zeta - \delta)^m(s, t) &= \sum_{s \preceq u \preceq t} (\zeta - \delta)(s, u) (\zeta - \delta)^{m-1}(u, t) \\ &= \dots = \sum_{s \prec u_1 \prec \dots \prec u_{m-1} \prec t} 1 \end{aligned}$$

, as $(\zeta - \delta)(s, u) = 1$ iff $s \prec u$. $\square$

Corollary 14.11. $(\zeta - \delta)^n = 0$ if $|P| \geq n$.

Theorem 14.12. $\zeta^m(s, t)$ is a polynomial in m of degree $\leq n$ (the degree is the length of the longest chain of the interval $[s, t]$).

Proof. Notice that $\zeta = (\zeta - \delta) + \delta$. Let $\zeta = A$, $\zeta - \delta = B$, $\delta = C$, then we have

$$\begin{aligned} \zeta^m(s, t) &= (A^m)_{st} \\ &= ((B + C)^m)_{st} \\ &= \left(\sum_{i=0}^m \binom{m}{i} B^i \right)_{st} \\ &= \left(\sum_{i=0}^m \frac{m(m-1)\dots(m-i+1)}{i!} B^i \right)_{st} \end{aligned}$$

, which is a polynomial of m of the degree we claimed. $\square$

Definition 14.13. The **order polynomial** $\Omega(P, m)$ is defined as the number of order preserving maps $f : P \rightarrow [m]$.

Remark 14.14. $\Omega(P, m) = \# \text{ chains from } \emptyset \text{ to } P \text{ in } J(P) \text{ of length } m-1$, which is therefore $\zeta_{J(P)}^m(\emptyset, P) \in \mathbb{Q}[m]$.

The degree of $\Omega(P, m)$ is $n = |P|$ and the leading term is $\frac{e(P)}{n!}$.

Conjecture 14.15. $\frac{\Omega(P, m)}{m^n} \geq \frac{\Omega(P, m+1)}{(m+1)^n}$.

Definition 14.16. The **Mobius function** $\mu \in I(P)$ s.t. $M(\mu) = M(\zeta)^{-1}$.

Theorem 14.17.

$$\mu(s, t) = \begin{cases} 1 & s = t \\ -\sum_{s \prec u \preceq t} \mu(u, t) & s \prec t \end{cases}$$

Proof. Notice that we have $M(\mu) = M(\zeta)^{-1}$ so $M(\mu) \cdot M(\zeta) = I = M(\delta) = M(\zeta) \cdot M(\mu)$.

Therefore,

$$\sum_{s \preceq u \preceq t} \mu(s, u) \zeta(u, t) = \delta_{s,t} = \begin{cases} 1 & s = t \\ 0 & \text{otherwise} \end{cases}.$$

Notice that accordingly, $\sum_{s \preceq u \preceq t} \mu(s, u) = 0$ and $\sum_{s \preceq u \preceq t} \mu(u, t) = 0$. Therefore, $\mu(s, t) = 0$ if $s = t$ and $\mu(s, t) = -\sum_{s \prec u \preceq t} \mu(s, u)$ if $s \neq t$. □

Example 14.18. If $P = C_n$, then

$$M(\mu) = M(\zeta)^{-1} = \begin{bmatrix} 1 & -1 & 0 & \dots & 0 \\ 0 & 1 & -1 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

Therefore, in this case,

$$\mu(i, j) = \begin{cases} 1 & i = j \\ -1 & i \prec j \\ 0 & \text{otherwise} \end{cases}.$$

Theorem 14.19. Let $f, g : P \rightarrow \mathbb{C}$. If $g(s) = \sum_{t \preceq s} f(t)$, for any $s \in P$, then $f(t) = \sum_{u \preceq t} \mu_P(u, t) g(u)$.

Proof. Notice that we have

$$[\dots f(t) \dots] M(\zeta) = [\dots f(t) \dots] \cdot \begin{bmatrix} 1 & \dots & \dots & \dots & \dots \\ 0 & 1 & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} = [\dots g(s) \dots].$$

Therefore, $[\dots f(t) \dots] M(\zeta) = [\dots g(s) \dots]$ so $[\dots f(t) \dots] = [\dots g(s) \dots] M(\mu)$. □

Corollary 14.20. $\zeta_{P \times Q}((s, s'), (t, t')) = \zeta_P(s, t) \zeta_Q(s', t')$.

$$\mu_{P \times Q}((s, s'), (t, t')) = \mu_P(s, t) \mu_Q(s', t').$$

Corollary 14.21. Notice that $\mu_{[2]}(a, b) = 1$ if $a = b$ and -1 if $a < b$.

Therefore, $\mu_{B_n}(S, T) = \prod \mu_{[2]}(a_i, b_i)$, where $S = \{a_1, \dots, a_n\}$, $T = \{b_1, \dots, b_n\}$.

Example 14.22. Let $P = B_n$, with $A_1, \dots, A_n$ fixed subsets of B_n .

Let $g(\{i_1, \dots, i_k\}) = |A_{i_1} \cap \dots \cap A_{i_k}|$ and $f(\{i_1, \dots, i_k\}) = |x \in A_i \Leftrightarrow i \in \{i_1, \dots, i_k\}|$. In other words, $g(\{i_1, \dots, i_k\})$ is the number of elements in the intersection of these k subsets, and $f(\{i_1, \dots, i_k\})$ is the number of elements **exactly** in these k subsets.

Then we have $g(S) = \sum_{S \subset T} f(T)$, so $f(T) = \sum_{T \subset S} (-1)^{|S-T|} g(S)$.

Example 14.23. Let $D_n = \{m, m \mid n\}$, with $a \preceq b$ iff $a \mid b$.

Let $n = p_1^{n_1} \dots p_k^{n_k}$, $a = p_1^{\alpha_1} \dots p_k^{\alpha_k}$, $b = p_1^{\beta_1} \dots p_k^{\beta_k}$. We then have $a \preceq b \Leftrightarrow \alpha_i \leq \beta_i$ for every i .

Therefore, each $a \in D_n$ is represented by $(\alpha_1, \dots, \alpha_k)$ and $D_n \cong C_{n_1} \times \dots \times C_{n_k}$.

Notice that we then have

$$\zeta(a, b) = \begin{cases} 1 & i \mid j \\ 0 & \text{otherwise} \end{cases}.$$

and $\mu(a, b) = \mu_{C_{n_1}}(\alpha_1, \beta_1) \cdot \dots \cdot \mu_{C_{n_k}}(\alpha_k, \beta_k)$.

$$\mu(a, b) = \mu_{C_{n_1}}(\alpha_1, \beta_1) \cdot \dots \cdot \mu_{C_{n_k}}(\alpha_k, \beta_k) = \begin{cases} 0 & \exists i \text{ s.t. } \alpha_i < \beta_i - 1 \\ (-1)^j & j = \#i \text{ s.t. } \alpha_i = \beta_i - 1 \end{cases}.$$

Notice that by this definition, $\mu_{D_n} = \mu(\frac{b}{a})$, which equals 0 if $\frac{b}{a}$ is not square-free and $(-1)^p$ where p is the number of prime factors of $\frac{b}{a}$.

Corollary 14.24. If $g(m) = \sum_{d \mid m} f(d)$, then $f(n) = \sum_{d \mid n} g(d) \cdot \mu(\frac{n}{d})$.

Definition 14.25. The **Riemann-Zeta function** $\zeta(s) = \sum_{n \leq s} \frac{1}{n^s}$ for $s > 1$, s.t. $\mu(x) = \sum_{n \leq s} \mu(n)$.

Theorem 14.26. $\mu = \sum_{k=0}^{n-1} (-1)^k (\zeta - \delta)^k$ if $(\zeta - \delta)^n = 0$.

Proof. We have

$$\begin{aligned} \mu(s, t) &= \zeta^{-1}(s, t) \\ &= (I + (\zeta - \delta))_{st}^{-1} \\ &= (I + N)_{st}^{-1} \\ &= (I - N + N^2 - \dots \pm N^{n-1})_{st} \end{aligned}$$

as we have $(I + N)(I - N + N^2 - \dots) = I$. □

Corollary 14.27. $\mu(s, t) = \delta_{st} - C_1 + C_2 - \dots$ where $C_k = \#\{k\text{-chains } s \prec u_1 \prec \dots \prec u_{k-1} \prec t\}$.

Proof. This comes directly from the above theorem. □

15 Graph

Definition 15.1. A **graph** is a set $G = (V, E)$ where $E \subset V \times V$.

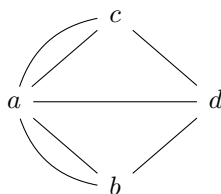
G is **simple** if $\nexists (v, v) \in E$ and there is at most one edge between any 2 vertices.

G is **connected** if for any $v, w \in V$, $\exists$ a path $v = v_0 \rightarrow v_1 \rightarrow \dots \rightarrow v_n = w$ with $(v_i, v_{i+1}) \in E$.

Definition 15.2. The **degree** of a vertex v , $\deg(v)$, is the number of edges from/to v .

Definition 15.3. A **Eulerian tour** is a path $v = v_0 \rightarrow v_1 \rightarrow \dots \rightarrow v_n = v$ with each edge in G appearing exactly once.

Example 15.4 (Seven Bridges). The following graph



has no Eulerian tour.

Theorem 15.5. A connected graph G has a Eulerian tour iff its vertices all have even degrees.

Proof. Suppose that there is an Eulerian tour, then the degree of every vertex is clearly even, as we always have a pair of paths, one into it and one out from it, for every vertex v_i in our path. (In particular, we consider $v_0 = v_n$).

Suppose that every vertex of G has an even degree, we prove the existence of an Eulerian tour by induction on $|E|$.

When $|E| = 1$ or 2 , the claim trivially holds.

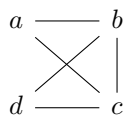
Suppose that the claim holds for $|E| \leq m$, then for $|E| = m + 1$, we start from an arbitrary v_0 and try to form an Eulerian tour. Suppose that j is the smallest such index with $v_j = v_i$ for some $i < j$. We then have a cycle $v_i \rightarrow v_{i+1} \rightarrow \dots \rightarrow v_j$.

Now, suppose that $G' = G \setminus \{(v_i, v_{i+1}), \dots, (v_{j-1}, v_j)\}$ can be expressed as the disjoint union of its components $G_1 \sqcup \dots \sqcup G_n$ and there is an Eulerian tour on each component by our induction assumption.

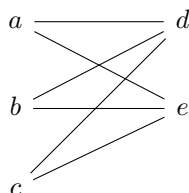
We clearly have a v_k in each G_i , where $v_k \in \{v_i, \dots, v_j\}$ to make the graph connected, and there is a (unique) edge between v_k and v_{k+1} for every k . Therefore, we are able to construct an Eulerian tour by concatenating the Eulerian tours on every G_i with an edge $v_k \rightarrow v_{k+1}$, and the claim is therefore proved. $\square$

Definition 15.6. A **Hamiltonian cycle** is a path $v = v_0 \rightarrow \dots \rightarrow v_n = v$ s.t. each vertex (except v) appears exactly once with $(v_i, v_{i+1}) \in E$ for any i .

Example 15.7. The graph below has a Hamiltonian cycle:



The graph below has no Hamiltonian cycle:



Conjecture 15.8. *Given a graph G , whether or not G has a Hamiltonian cycle is NP-complete.*

Theorem 15.9. *Suppose G is a simple graph, $|V| = n$, s.t. for any $(u, v) \notin E$, $\deg(u) + \deg(v) \geq n$, then G has a Hamiltonian cycle.*

Proof. Statement is clearly true for K_n .

Suppose $\exists K \subset K_n$ a graph satisfying the condition without a Hamiltonian cycle, let G be one of them with the maximal number of edges.

Thus, for any $(u, v) \notin E_G$, $G \cup \{(u, v)\}$ has a Hamiltonian cycle. Moreover, it is clear that (u, v) is contained in it.

Therefore, there exists a path in G with the form

$$u \rightarrow v_1 \rightarrow \dots \rightarrow v_{n-2} \rightarrow v,$$

such that each vertex is visited exactly once.

We know that $\deg(u) + \deg(v) \geq n$. Consider the following set of neighbors of u and v :

$$\begin{aligned} N_u &= \{v_1, v_{i_1}, \dots, v_{i_k}\} \text{ (thus } \deg(u) = k + 1\text{).} \\ N_v &= \{v_{n-2}, v_{j_1}, \dots, v_{j_m}\} \text{ (thus } \deg(v) = m + 1\text{).} \end{aligned}$$

We thus have $k + 1 + m + 1 \geq n$.

Consider the set $N_u - 1 = \{v_{i_1-1}, \dots, v_{i_k-1}\}$. We then have $|N_v| + |N_u - 1| = k + m + 1 \geq n - 1$.

Therefore, there is an element v_r s.t. $v_r \in N_v \cap (N_u - 1)$ by Pigeonhole's principle.

We thus have $(v_r, v) \in E$, $(u, v_{r+1}) \in E$.

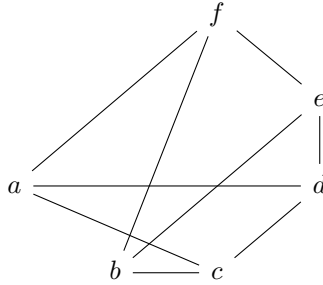
Therefore, we have the following Hamiltonian cycle:

$$u \rightarrow v_1 \rightarrow \dots \rightarrow v_r \rightarrow v \rightarrow v_{n-2} \rightarrow \dots \rightarrow v_{r+1} \rightarrow u,$$

contradicting the fact that there is no Hamiltonian cycle in G .

Therefore, any graph satisfying the conditions has a Hamiltonian cycle. $\square$

Example 15.10. *We can claim that the following graph has a Hamiltonian cycle without explicitly finding it:*



16 Proper Coloring of Graphs

Definition 16.1. A **proper coloring** of a simple graph G is a map $\mathcal{K} : [n] \rightarrow V$ s.t. $(u, v) \in E \Rightarrow f(u) \neq f(v)$.

Definition 16.2. The minimal number n to properly color a graph G is called the **chromatic number** of G , denoted as $\chi(G)$.

Definition 16.3. G is **bipartite** if $V = V_1 \sqcup V_2$ s.t. for any $(u, v) \in E$, $u \in V_1, v \in V_2$ or $u \in V_2, v \in V_1$ (i.e., $E \subset V_1 \times V_2$).

Corollary 16.4. $\chi(G) \leq 2$ iff G is bipartite.

Corollary 16.5. If G has an odd cycle then $\chi(G) \geq 3$.

Theorem 16.6. If G has only even cycles then $\chi(G) \leq 2$.

Proof. Fix $v \in V$. Define $d(u)$ to be the minimal k s.t. there is a path $v \rightarrow v_1 \rightarrow \dots \rightarrow v_k = u$.

We construct a coloring for G by letting $\mathcal{K}(u) = i$ if $d(u) \equiv i \pmod{2}$.

If $\mathcal{K}$ is not proper, then there exist $u_1, u_2 \in V$, $(u_1, u_2) \in E$, and $\mathcal{K}(u_1) = \mathcal{K}(u_2)$.

We thus have a cycle

$$v \rightarrow v_1 \rightarrow \dots \rightarrow v_{d(u_1)} (= u_1) \rightarrow v'_{d(u_2)} (= u_2) \rightarrow \dots \rightarrow v'_1 \rightarrow v.$$

which is a cycle of length $d(u_1) + d(u_2) + 1$ which is odd as $d(u_1)$ and $d(u_2)$ have the same parity, contradicting the restriction on G . $\square$

Definition 16.7. The **chromatic polynomial** of G , $t_G(k)$, is defined to be the number of proper coloring of G by k colors.

Example 16.8. Let $G = K_n$, then $\chi(G) = n$ and $t_G(n) = n!$.

More generally,

$$t_G(k) = \begin{cases} 0 & k < n \\ k(k-1)\dots(k-n+1) & k \geq n \end{cases}.$$

The discrete space of n points, $\cdot^n$, has $t_{\cdot^n}(k) = k^n$.

Theorem 16.9. $t_G(k) = t_{G-\{(u,v)\}}(k) - t_{G/(u,v)}(k)$, where $G/(u,v)$ is the graph obtained by contracting (u,v) into a point.

Proof. Notice that $t_{G-\{(u,v)\}}(k)$ is the number of proper colorings with $c(u) \neq c(v)$ or proper colorings with $c(u) = c(v)$.

The former corresponds to the number of proper coloring of G , $t_G(k)$, and the latter corresponds to the number of proper coloring of G after contracting (u,v) , $t_{G/(u,v)}(k)$, so the equality follows. $\square$

Proposition 16.10. $t_G(k)$ is a polynomial with integer coefficients of k ($t_G(k) \in \mathbb{Z}[k]$) of degree $|V_G|$.

Proof. We prove the statement by induction on $|E_G|$.

If $|E_G| = 0$, then $t_G(k) = k^n$.

If $|E_G| = 1$, then $t_G(k) = k^{n-1}(k-1)$.

Now suppose that the claim holds for $|E_G| = m$, then for $|E_G| = m+1$, we have $t_G(k) = t_{G-\{(u,v)\}}(k) - t_{G/(u,v)}(k)$ by the above proposition.

By the induction assumption, $t_{G-\{(u,v)\}}(k)$ is a polynomial of degree $|V_G|$ and $t_{G/(u,v)}(k)$ is a polynomial of degree $|V_G| - 1$. Therefore, $t_G(k)$ is also a polynomial of degree $|V_G|$. $\square$

Remark 16.11. *The leading coefficient of $t_G(k)$ is always 1, as can be proved by induction.*

Example 16.12. (i) *Let P_n be a straight line with n points on it, then $t_{P_n}(k) = k(k-1)^{n-1}$. In particular, $\chi(P_n) = 2$ and $k = 0, 1$ are both roots.*

(ii) *Let G be a square, then $t_G(k) = t_{P_4}(k) - t_{K_3}(k) = k(k-1)^3 - k(k-1)(k-2) = k(k-1)(k^2 - 3k + 3)$. In particular, $\chi(G) = 2$ and $k = 0, 1$ are both roots.*

(iii) *Let H be a polygon, then $t_H(k) = t_{P_5}(k) - t_G(k) = k(k-1)^4 - k(k-1)(k^2 - 3k + 3) = k(k-1)((k-1)^3 - k^2 + 3k - 3)$. In particular, $\chi(H) = 3$ and $k = 0, 1, 2$ are all roots.*

Theorem 16.13. *If $\chi(G) = m$, then $t_G(k) = k(k-1)\dots(k-m+1)p(k)$ with $p(k)$ a polynomial of k in $\mathbb{Z}[k]$ of degree $|V_G| - m$.*

17 Tree

Definition 17.1. A simple connected graph G on n vertices is a **tree** if it satisfies one of the following three conditions (it is not difficult to verify that they are equivalent):

- (i) G has no cycle.
- (ii) Removing any edge from G makes G disconnected.
- (iii) $|E_G| = n - 1$.

Corollary 17.2. Every tree has at least 2 leaves.

Proof. We have $\sum \deg(v_i) = 2|E_G| = 2n - 2$. As G is connected, $\deg(v) \geq 1$ for any $v \in G$. If there is only one leaf in G , then $\sum \deg(v_i) \geq 2(n - 1) + 1 > 2n - 2$, leading to contradiction. $\square$

Remark 17.3. # leaves in $G \geq \max(\deg(v_i))$.

Theorem 17.4. $t_T(k) = k(k - 1)^{n-1}$, if T is a tree with n vertices.

Proof. Let T be a tree, u be a leaf.

We then have $t_T(k) = t_{T-\{(u,v)\}}(k) - t_{T/(u,v)}(k) = kt_{T'} - t_{T'} = (k - 1)t_{T'}$, where T' is a tree on $n - 1$ vertices.

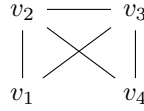
Therefore, we can prove the statement by induction. $\square$

Definition 17.5. Let G be a graph with $V_G = \{v_1, \dots, v_n\}$. We define

$$X_G(X_1, \dots, X_n) = \sum_{c: V_G \rightarrow [k]} X_{c(v_1)} \dots X_{c(v_n)},$$

where c is a proper coloring of G .

Example 17.6. Let G be the following graph



with the coloring $c(v_1) = 3$, $c(v_2) = 1$, $c(v_3) = 2$, $c(v_4) = 3$, then $X_{c_{v_1}} \dots X_{c_{v_4}} = X_1 X_2 X_3^2$.

Conjecture 17.7. Let T and T' be non-isomorphic trees, then $X_T(X_1, \dots, X_m) \neq X_{T'}(X_1, \dots, X_m)$.

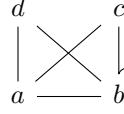
Remark 17.8. $X_G(1, 1, \dots, 1) = t_G(k)$.

18 Planar Graphs

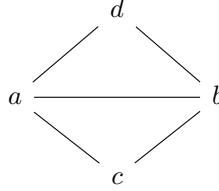
Definition 18.1. A graph G is **planar** iff it can be drawn on a plane without an intersection.

Example 18.2. The following graphs are planar:

(i) The graph below

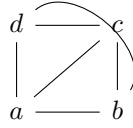


is isomorphic to



so it is planar.

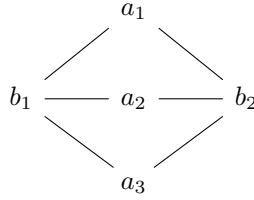
(ii) K_4 can be drawn as



so it is planar.

(iii) Any arbitrary tree is a planar graph by drawing its tree structure.

(iv) $K_{2,3}$, the complete bipartite graph between 2 and 3 vertices, can be drawn as



so it is planar.

Theorem 18.3 (Euler). If G is planar and connected and is drawn with $|F|$ faces, then $|V| + |F| = |E| + 2$.

Proof. Let $|V| = n$, we do induction on $|E|$.

When $|E| = n - 1$ (i.e., G is a tree), $|F| = 1$, and we have $n + 1 = (n - 1) + 2$ so the theorem holds.

Now suppose that the claim holds for $|E| \leq m$, then for $|E| = m + 1$, we have some edge $(u, v) \in E$ s.t. $G - (u, v)$ is still connected. Let $G' = G \setminus \{(u, v)\}$, then we have $|V_{G'}| + |F_{G'}| = |V_G| + |F_G| - 1 = |E_G| + 3 - 1 = |E_G| + 2$. $\square$

Definition 18.4. Given G a planar graph, we define the **dual graph** by setting up a vertex v at each face and an edge between two vertices (face) if there is an edge connecting the two faces.

The degree of a face in G , denoted by $\deg(f)$, is the number of edges from/to the vertex representing this face in the dual graph.

Corollary 18.5. $\sum_{f \in F} \#(e \in f) = \sum_{\substack{e \in E, \\ e \in f \in F}} (e, f) = \sum_{e \in E} \sum_{f \ni e} 1 \leq 2 \deg(E)$.

Theorem 18.6. K_5 is not planar.

Proof. Suppose that K_5 is planar.

We have $|V| + |F| = |E| + 2$, so $|F| = 7$. As $\sum \deg(f) \leq 2|E| = 20$, we have that $\deg(f) < 3$ for some face f .

However, we do not have self-loop or cycle of 2 vertices in K_5 , so $\deg(f) \geq 3$ for any f , contradicting the claim that K_5 is planar. $\square$

Theorem 18.7. $K_{3,3}$ is not planar.

Proof. Suppose that $K_{3,3}$ is planar.

We have $|V| + |F| = |E| + 2$, so $|F| = 11 - 6 = 5$. We also have that $\sum \deg(f) \leq 2|E| = 18$, so $\deg(f) < 4$ for some face f .

However, $\deg f \geq 4$ for any f since otherwise f is going to be a self-loop or cycle of 2 or 3 vertices, all of which does not exist in $K_{3,3}$. $\square$

Theorem 18.8 (Kuratowski). G is planar iff G has no subgraph H s.t. $H \sim K_5$ or $K_{3,3}$, where $G \sim H$ iff $G = H$ by contracting an edge of G or H .

Theorem 18.9 (4-color Theorem). Any planar graph can be properly colored by 4 colors (or less).

Corollary 18.10. Any planar graph can be properly colored by 5 colors (or less).

To prove the corollary, we prove the lemma below first:

Lemma 18.11. In every planar simple graph, there is a vertex v with degree at most 5.

Proof. We have that $\sum \deg(f) \leq 2|E|$. Additionally, $3|F| \leq \sum \deg(f)$, as G is simple so that every face has at least 3 edges surrounding it. We then have

$$\begin{aligned} 3|F| \leq 2|E| &\Rightarrow 3(|E| - |V| + 2) \leq 2|E| \\ &\Rightarrow 3|V| \geq |E| + 6 \\ &\Rightarrow \sum \deg(v) \leq 6|V| - 12 \\ &\Rightarrow \deg(v) < 6 \text{ for some } v \in V. \end{aligned}$$

$\square$

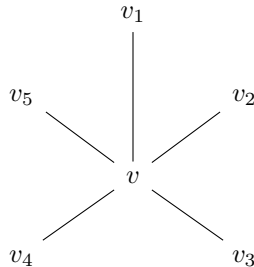
Now comes the proof for the corollary:

Proof. We prove the statement by induction on $n = |V|$.

When $n \leq 5$, $\chi(G) \leq 5$.

Suppose the claim holds for $|V| \leq n$, then for $|V| = n + 1$, if $\deg(v) < 5$ for some $v \in V$ then the claim is trivial, since $G \setminus \{v\}$ has a proper coloring of 5 colors.

Now, let $\deg(v) = 5$ and fix a proper coloring of $G \setminus \{v\}$. Suppose the graph at v is drawn below:



with v_5 having the color *green* and v_3 having the color *blue*.

We then want to flip the colors on the *green* – *blue* chain from v_5 . If it does not intersect v_3 then we are done. Otherwise, we are going to have a loop around v_1 (or v_2 and v_3).

Suppose that the loop is around v_4 , and v_4 has color *red*, v_1 has color *yellow*. We can then flip the *red* – *yellow* chain from v_4 and it will not intersect v_1 since G is planar. We therefore reach a proper color of v (with color *red*) and proves the claim. $\square$

19 Ramsey Theory

Theorem 19.1. *If we color K_6 with each edge either white or red, then there is a subgraph of K_6 isomorphic to K_3 formed by all white or red edges.*

Proof. Take $v_1 \in K_6$. At least 3 of the edges (v_1, v_i) are of the same color, call them $(v_1, v_{i_1}), (v_1, v_{i_2}),$ and (v_1, v_{i_3}) .

If any of (v_{i_j}, v_{i_k}) is of the same color as (v_1, v_{i_j}) , then (v_1, v_{i_j}, v_{i_k}) form such a K_3 .

Otherwise, $(v_{i_1}, v_{i_2}, v_{i_3})$ form such a K_3 . □

Definition 19.2. *Let $k, l \in \mathbb{N}$. The **Ramsey number** $R(k, l)$ is the minimal positive integer n s.t. if we color the edges in K_n in the color white or red, then there always exists either a K_k with all white edges or a K_l with all red edges.*

Corollary 19.3. $R(3, 3) = 6$.

Theorem 19.4. $R(k, l) \leq \binom{k+l-2}{k-1}$.

Proof. We want to show that $R(k, l) \leq R(k-1, l) + R(k, l-1)$ by induction on $k+l$.

Clearly $R(2, l) = l$ and $R(k, l) = R(l, k)$, and the claim holds when $k+l = 4$.

Now for $k+l$, let $n = n_1 + n_2$, where $n_1 = R(k-1, l)$ and $n_2 = R(k, l-1)$.

Let $G = \{v\} \cup K_{n-1}$. From v , there are at least n_1 white edges or n_2 red edges.

Assume that there are n_1 white edges from v mapped to n_1 vertices in K_{n-1} , which, with edges between them, form a subgraph $G_{n_1} \subset K_{n-1}$. We know that there is at least a K_{k-1} of all white edges or K_l of all red edges in G_{n_1} .

If it is the latter, we are done. If it is the former, $\{v\} \cup K_{k-1} = K_k$ so the claim holds.

Therefore, the claim holds for any $k+l$. We can then show the statement by proving $R(k, l) \leq \binom{k+l-2}{k-1}$ by induction, as we have

$$R(k, l) \leq R(k, l-1) + R(k-1, l) \leq \binom{k+l-3}{k-1} + \binom{k+l-3}{k-2} = \binom{k+l-2}{k-1}.$$

□

Theorem 19.5. $R(3, 4) = 9$.

Proof. We first show that there is either a K_3 with all edges white or a K_4 with all edges red in any coloring of K_9 .

Let $G = \{v, v_1, \dots, v_8\}$.

(1) Suppose that there are at least 4 edges, namely $(v, v_1), \dots, (v, v_4)$, from v that are white.

If any of (v_i, v_j) ($1 \leq i, j \leq 4$) is white, then (v, v_i, v_j) form a K_3 with all edges white.

Otherwise, if non of (v_i, v_j) is white, then $(v_1, \dots, v_4)$ form a K_4 with all edges red.

(2) Suppose that there are at least 6 edges from v that are red, namely $(v, v_1), \dots, (v, v_6)$.

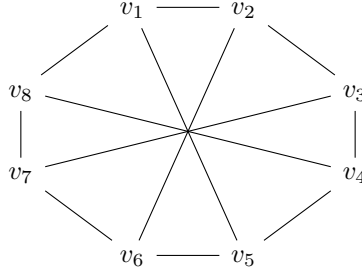
As $R(3, 3) = 6$, we know that there is a $K_3 \subset K_6 = (v_1, \dots, v_6)$ with all edge white or red.

If it is the former, then we are done. If it is the latter, then $\{v\} \cup K_3$ is a K_4 with all edges red.

(3) Suppose that all vertices have exactly 3 edges from them that are white. We want to show that this is impossible.

Notice that in such a G , there are in total $9 \times 3 = 27$ white edges, but the number apparently has to be even. Therefore, there does not exist such a graph G .

To show that $R(3, 4) > 8$, notice that there does not exist either a K_3 with all edges *white* (denoted by the lines drawn explicitly) or a K_4 with all edges *red* (the *red* edges are those not drawn explicitly) in the below coloring of K_8 :



□

Definition 19.6. $R(n_1, \dots, n_k)$ is defined as the minimal $n \in \mathbb{N}$ s.t. K_n in any coloring has a subgraph isomorphic to K_{n_1} with all edges in color 1, ..., or a subgraph isomorphic to K_{n_k} with all edges in color k .

Theorem 19.7. $R(n_1, \dots, n_k) \leq R(n_1 - 1, n_2, \dots, n_k) + R(n_1, n_2 - 1, \dots, n_k) + \dots + R(n_1, n_2, \dots, n_k - 1)$.

Corollary 19.8. $R(2, k, l) = \max(R(k, l), 1)$.

Corollary 19.9. Suppose that we color every point in $\mathbb{R}^2$ by two colors, then

- (i) There exists $x, y \in \mathbb{R}^2$ with distance 1, s.t. x and y are of the same color.
- (ii) There exists a right triangle on $\mathbb{R}^2$ with degree of its angles $30^\circ, 60^\circ, 90^\circ$, s.t. all of its vertices are of the same color.

Proof. (i) Consider any equilateral triangle with side length 1.

- (ii) Suppose that the claim is false. Let A, B be two points in $\mathbb{R}^2$ of the same color and distance 1.

Let C, D be two points s.t. $ABCD$ form a square with diagonal length 1. Let E, F be two points s.t. AEB, BFA are triangles with $\angle AEB = \angle BFA = 90^\circ$, $\angle BAE = \angle ABF = 30^\circ$.

Notice that C, D, E, F are of different color compared to A and B , so they are in the same color. The triangles CEF and DEF then satisfy the condition, leading to a contradiction.

□

Theorem 19.10. There exists 4 monochromatic points in $\mathbb{Z} \times \mathbb{Z}$ forming a rectangle for any coloring with a finite number of colors.

Proof. Suppose that we have n colors. We then know that there exists at least 2 monochromatic points among $(1, 1), \dots, (1, n + 1)$.

Since the number of permutations of n colors on $n + 1$ points is finite (n^{n+1}), we then are guaranteed to have 2 columns of $n + 1$ points having exactly the same coloring among $n^{n+1} + 1$ columns of $n + 1$ points.

Therefore, picking the monochromatic 2 points in each of these 2 columns we form a monochromatic rectangle. □

Theorem 19.11 (van der Waerden). For any $r, l \in \mathbb{N}$, there exists $N \in \mathbb{N}$ s.t. if we color $[N]$ in r colors, then there exists a monochromatic sequence of l integers $a_1, \dots, a_l$ s.t. $a_{i+1} - a_i = c$ for some c (i.e., they form an arithmetic sequence).

N is called the **van der Waerden number**, denoted as $W(r, l)$. It is proved that $W(r, k) \leq 2^{r \cdot 2^{k+9}}$.

Corollary 19.12. $W(2, 3) = 9$.

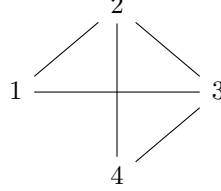
Theorem 19.13 (Green-Tao). The sequence of prime numbers have arbitrarily long arithmetic sequence.

20 Spanning Tree in Graph

Definition 20.1. Given a graph $G = (V, E)$, its **adjacency matrix** A is defined as

$$A_{ij} = \begin{cases} 1 & (i, j) \in E \\ 0 & (i, j) \notin E \end{cases}.$$

Example 20.2. Consider the graph G as below



then its adjacency matrix A would be

$$\begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}.$$

Notice that the adjacency matrix is always symmetric.

Theorem 20.3. The number of paths of length m from i to j is $(A^m)_{ij}$.

Proof. We have

$$\begin{aligned} A_m)_{ij} &= \sum_{k=1}^n A_{ik} A_{kj} \\ &= \sum_{k_1=1}^n \sum_{k_2=1}^n \dots \sum_{k_{m-1}=1}^n A_{ik_1} A_{k_1 k_2} \dots A_{k_{m-1} j}. \end{aligned}$$

Notice that the inner product is 1 if $i \rightarrow k_1 \rightarrow \dots \rightarrow k_{m-1} \rightarrow j$ is a valid path (and 0 if it is not). $\square$

Definition 20.4. The **Laplacian matrix** is defined as

$$\begin{bmatrix} d_1 & 0 & \dots & 0 \\ 0 & d_2 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & \dots & 0 & d_n \end{bmatrix} - A,$$

where $d_i = \deg(i)$.

Example 20.5. For the above G , its Laplacian matrix L would be

$$\begin{bmatrix} 2 & -1 & -1 & 0 \\ -1 & 3 & -1 & -1 \\ -1 & -1 & 3 & -1 \\ 0 & -1 & -1 & 2 \end{bmatrix}.$$

Corollary 20.6. We have $\sum_{j=1}^n A_{ij} = 0$ for any $1 \leq i \leq n$. Therefore, $\det(L) = 0$ as $L \cdot [1, \dots, 1]^T = 0$.

Definition 20.7. A **spanning tree** of $G = (V, E)$ is a subgraph T with $V_T = V_G$, $E_T \subset E_G$, and T being a tree.

Theorem 20.8 (Matrix-Tree). $\det(L_i)$ is the number of spanning trees on G , where L_i is L with the i th row and column removed.

Lemma 20.9 (Cayley-Hamilton). $\det(Ix - A) = (x - \lambda_1) \dots (x - \lambda_n)$, where the λ 's are the eigenvalues (not necessarily distinct) of $Ix - A$.

Example 20.10. Consider K_n . Its Laplacian matrix L is the $n \times n$ matrix with diagonal filled by $n - 1$ and all other entries -1 . Consider L_n which is clearly of the same form but without the last row and column.

$\det(L_n) = \det(In - A) = (n - \lambda_1) \dots (n - \lambda_n) = (n - 0)^{n-2} (n - (n - 1)) = n^{n-2}$, where A is the $(n - 1) \times (n - 1)$ matrix filled by 1's.

Theorem 20.11 (Cayley). The number of labeled trees on n vertices is n^{n-2} .

21 Matching

Definition 21.1. Given a graph $G = (V, E)$, a **matching** M is a set of edges $\{(u_i, v_i) \mid \{u_1, \dots, u_k, v_1, \dots, v_k\} \text{ all distinct}\}$.
A matching is **perfect** if $\{u_1, \dots, u_k, v_1, \dots, v_k\} = V$.

Theorem 21.2 (Hall's Marriage Theorem). A bipartite graph G with $E \subset V_1 \times V_2$ has a matching covering V_1 iff

$$|N(S)| \geq |S|$$

for all $S \subset V_1$, where $N(S) = \{v \in V_2, (s, v) \in E \text{ for some } s \in S\}$.

Proof. Let $V_1 = \{1, \dots, n\}$, $V_2 = \{w_1, \dots, w_m\}$.

(1) Suppose that there is a matching covering V_1 , let $(i, w_i) \in M$.

Then we have $N(S) \supset \{w_i \mid i \in S\}$, so $|N(S)| \geq |S|$ for any $S \subset V_1$.

(2) On the other hand, suppose that $|N(S)| \geq |S|$ for any $S \subset V_1$.

If $|N(S)| > |S|$ for any $S \subset V_1$, $S \neq V_1$ or $\emptyset$, then we do induction on $|V_1| = n$.

Suppose that $(n, w_n) \in E$. Let $G' = G \setminus \{n, w_n\}$, $V_1' = V_1 \setminus \{n\}$.

Then for any $S \subset V_1'$, we have

$$\begin{aligned} |N(S)|_{G'} &\geq |N(S) \setminus \{w_n\}| \\ &\geq |N(S)| - 1 \\ &\geq |S| \end{aligned}$$

so there is a matching M' covering $V_1|_{G'}$. Therefore, $M = M' \cup \{(n, w_n)\}$ is a matching covering V_1 .

On the other hand, suppose that there is some $S \subset V_1$, $S \neq V_1$, s.t. $|N(S)| = |S|$. We still do induction on $|V_1|$.

Let $G' = S \cup N(S)$, $G'' = (V_1 \setminus S) \cup (V_2 \setminus N(S))$.

If $S_1 \subset S$, then $N(S_1) \subset N(S)$, so $|N(S_1)|_{G'} = |N(S_1)| \geq |S_1|$. Therefore, by induction, there is a matching $M' : S \rightarrow N(S)$.

On the other hand, let $T = V_1 - S$, $S_2 \subset T$.

Then we have

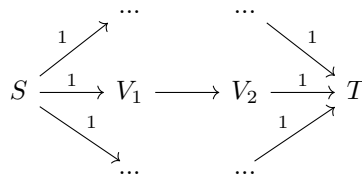
$$\begin{aligned} |S| + |S_2| &= |S \cup S_2| \\ &\leq |N(S \cup S_2)| \\ &= |N(S)| + |N(S_2)|_{G''} \end{aligned}$$

so $|N(S_2)|_{G''} \geq |S_2|$, so by induction, there exists a matching $M'' : T \rightarrow N(T) \setminus G'$.

We thus have $M = M' \cup M''$ a matching covering V_1 .

□

Corollary 21.3. The maximal flow from S to T in the graph below is exactly the maximal number of $|M|$, where $M : V_1 \rightarrow V_2$ is a matching.



Definition 21.4. Given a $n \times n$ matrix A , we have

$$\det(A) = \sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_{i=1}^n A_{i\sigma_i},$$

$$\text{perm}(A) = \sum_{\sigma \in S_n} \prod_{i=1}^n A_{i\sigma_i}.$$

Theorem 21.5. *If a graph G can be written as $V_1 \cup V_2$ with $E \subset V_1 \times V_2$ and $|V_1| = |V_2|$, then $\text{perm}(A)$ is the number of perfect matchings of G .*

Proof. If and only if $M = \{(i, w_j)\}$ be a perfect matching can we have $\prod A_{i\sigma_i} = 1$, where σ sends i to j for each $i \in [n]$. Otherwise, $\prod A_{i\sigma_i} = 0$ (as either $A_{i\sigma_i} = 0$ for some i or σ represents a perfect matching).

In other words, we have

$$\prod A_{i\sigma_i} = \begin{cases} 1 & (i, w_{\sigma_i}) \in E \text{ so } M = \{(1, w_{\sigma_1}), \dots, (n, w_{\sigma_n})\} \text{ is a perfect matching} \\ 0 & \text{otherwise} \end{cases}$$

□

Theorem 21.6 (FKT Algorithm). *Let G be a planar bipartite graph, T be a spanning tree of G , and we arbitrarily choose a orientation for the edges of T .*

Let T' be the dual graph where (f_i, f_j) is an edge in T' iff the edge between them is not in T .

Now we give each edge in $E \setminus T$ an orientation s.t. the number of edges in the cycle oriented clockwise is odd (we can always do this by recurring from the edges in $E \setminus T$ between leaves of T' and their parents).

Now we define A_{ij} to be

$$A_{ij} = \begin{cases} 1 & i \rightarrow j \\ -1 & j \rightarrow i \\ 0 & \text{otherwise} \end{cases}$$

then the number of perfect matching of G is $|Pf(A)|$, and $|Pf(A)|^2 = \det(A)$, where $Pf(A) = \sum_{\sigma} \text{sgn}(\sigma) \prod A_{i\sigma_i}$ and σ is a perfect matching of G .